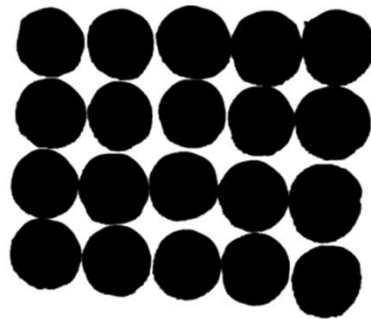


Filters and Spatial Operations

Introductory Course to Image Analysis in Life Science

LECLERC SIMON

General Image Analysis Workflow



Number of
tomatoes: 20
Average tomato
diameter: 6cm

Image

Images are made
of pixels –
intensity values at
a spatial position

Segmentation

Derived images
(e.g. denoised
image, edge
enhancement)

Objects

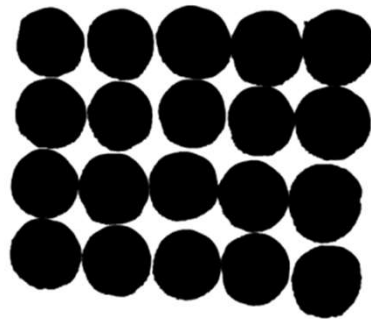
Pixels are grouped
into recognisable
regions/objects

Measurements

Objects are
measured for their
shape, size, counts
etc.

Core Problem: Object Segmentation

General Image Analysis Workflow



Number of
tomatoes: 20
Average tomato
diameter: 6cm

Image

Segmentation

Objects

Measurements

Image processing

Applying mathematical operations to an input image to enhance / suppress features in an output image.

Segmentation

Partitioning pixels into identifiable groups/regions

Quantification

Measuring objects for their size, shapes, counts, intensities in specific channels

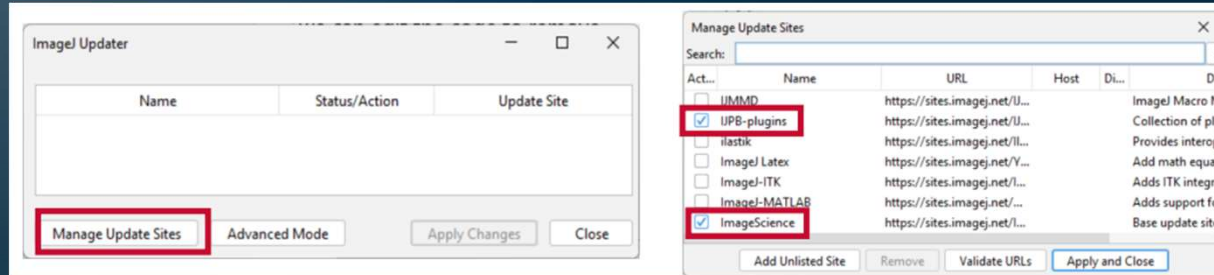
Contents

- Global thresholding
 - Manual thresholding
 - Automated thresholding
- Image filters
 - Denoising
 - Feature enhancement
 - Background removal
- Adaptive segmentation
 - Local adaptive threshold
 - Local minima/maxima + watershed
 - Machine learning pixel classification
- Binary mask processing
 - Object Labelling
 - Distance Transform Watershed
 - Morphological filters
 - Binary Mask Operations
 - Skeletonization Image filters
- Quantification
 - Regions
 - Fibre/Networks
 - Points
- 3D Analysis

Fiji Is Just ImageJ - Fiji

- Fiji comes with a lot of plugins installed, but you can always access more by installing them as .jar files. An easy way to do so is with Update Sites
- For this class we will need to install the FeatureJ and MorpholibJ plugins for our exercises, which are available from the ImageScience and IJPB-plugins update sites respectively

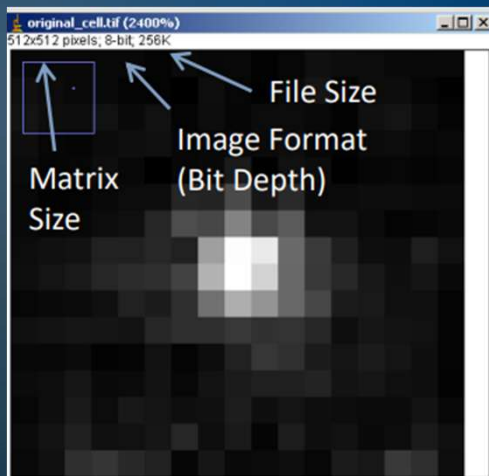
1. Use **Help > Update...** , then **Manage Update Sites**



2. Select IJPB-Plugins and ImageScience, click 'Apply and Close', and then 'Apply Changes'. Restart Fiji and you should see them listed under 'Plugins'

Digital Image Formation

A digital image is a matrix of scalar numbers:

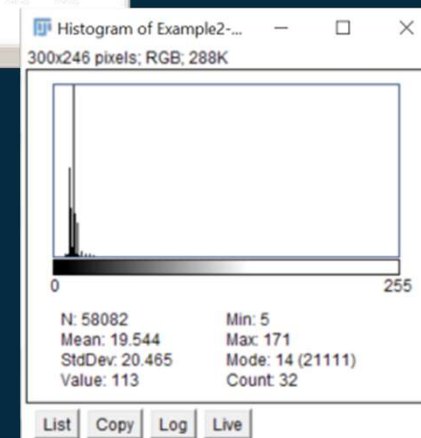


The screenshot shows a WordPad window titled 'original_cell_ci_roi_crop.txt'. It contains a 13x12 matrix of scalar values.

19	22	25	19	0	0	10	37	0	15	20	
29	16	8	11	16	16	13	12	0	13	17	
14	5	1	11	29	36	23	5	5	11	13	
0	5	19	33	54	70	62	43	19	12	13	
12	29	46	70	115	152	140	103	38	19	18	
58	56	57	92	176	238	210	140	61	30	28	
63	58	50	82	175	249	217	135	82	42	38	
34	42	40	61	138	208	186	115	95	50	45	
17	38	51	51	60	78	78	64	42	29	29	
8	21	22	15	26	53	67	63	49	37	37	
7	18	17	6	14	37	49	43	42	29	29	
13	31	38	30	31	40	34	17	25	12	12	

There is a distribution of intensity values in the image

Analyze > Histogram (Ctrl + H)

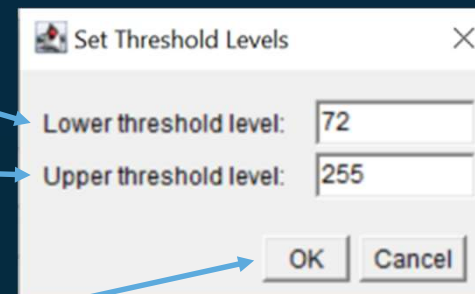
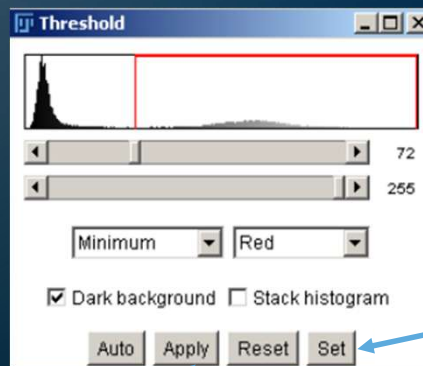


Manual Pixel Classification

Pixels can be classified as Foreground/Background based solely on their intensity

Image > Adjust > Threshold (Shift+T)

- Thresholds T1 and T2 can be adjusted **manually**
- Pixels with intensity within [T1, T2] classified as foreground



Apply Threshold

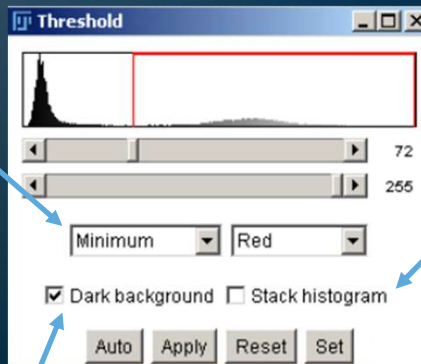
Manual Pixel Classification

Pixels can be classified as Foreground/Background based solely on their intensity

Image > Adjust > Threshold (Shift+T)

- Thresholds T1 and T2 can be adjusted **manually**
- Pixels with intensity within [T1, T2] classified as foreground

Algorithm



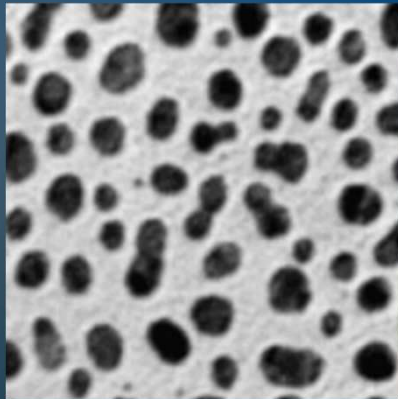
Tick when
working
with stack

Parameters

- Safest: adjust threshold manually for each image
- Acceptable: Use algorithm adapted to the data, beware of heterogeneous datasets!
- Misconception: Automated methods are more accurate and robust

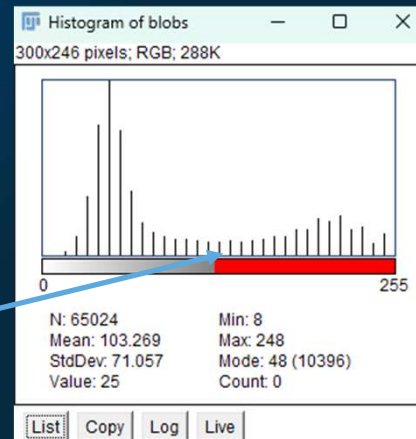
Minimum Algorithm

File > Open Samples > Blobs

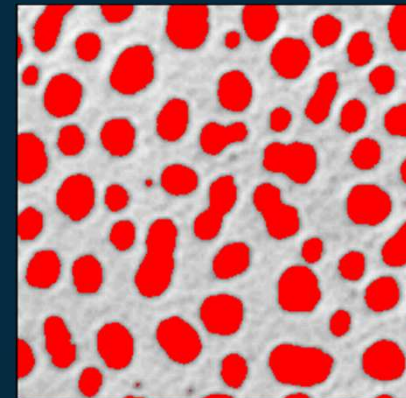


Intensity
Histogram

Bimodal
(2 peaks)



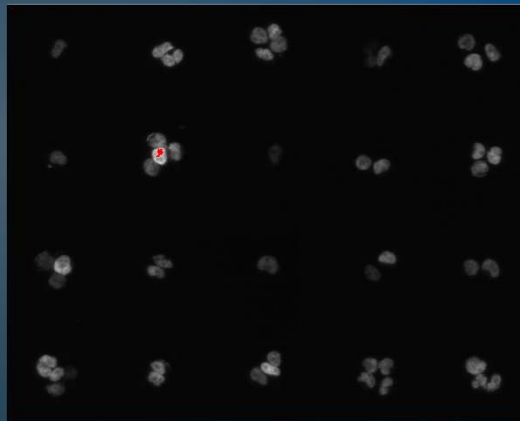
Selected Threshold



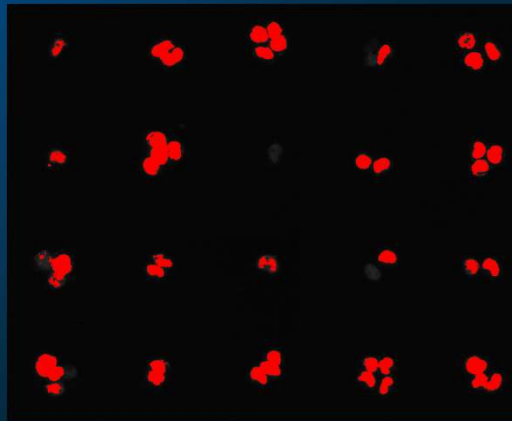
1. Smooth histogram until two **Significant Maxima** are found
 2. Set T1 or T2 midway between the two maxima
- Assumption: Two underlying balanced and uniform intensity population

Otsu's Algorithm

Example1 – DAPI.tif

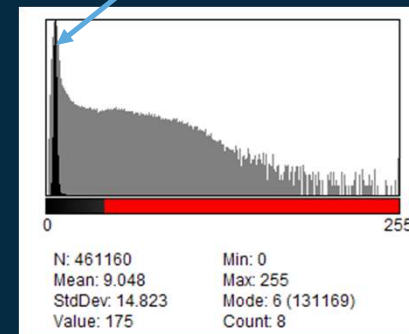


Minimum Algo (red)



Otsu's Algo (red)

Only one significant Maximum



Intensity Histogram,
log in grey

1. For all possible T_1 , compute the **standard intensity deviations** for the pixel in Foreground and Background classes
2. Select T_1 **minimizing** the overall standard deviation (weighted average)

Assumption: Two underlying uniform intensity population

Other Global Thresholding Algorithms

How about other methods?

Image > Adjust > Auto Threshold

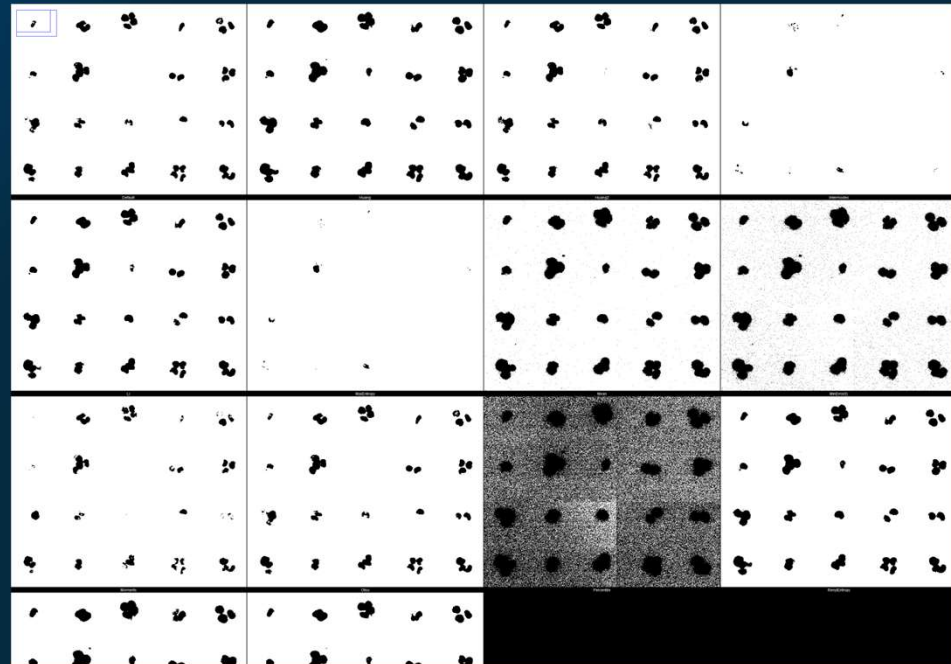
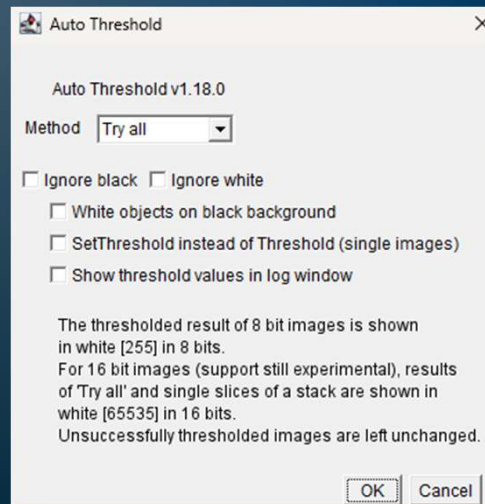
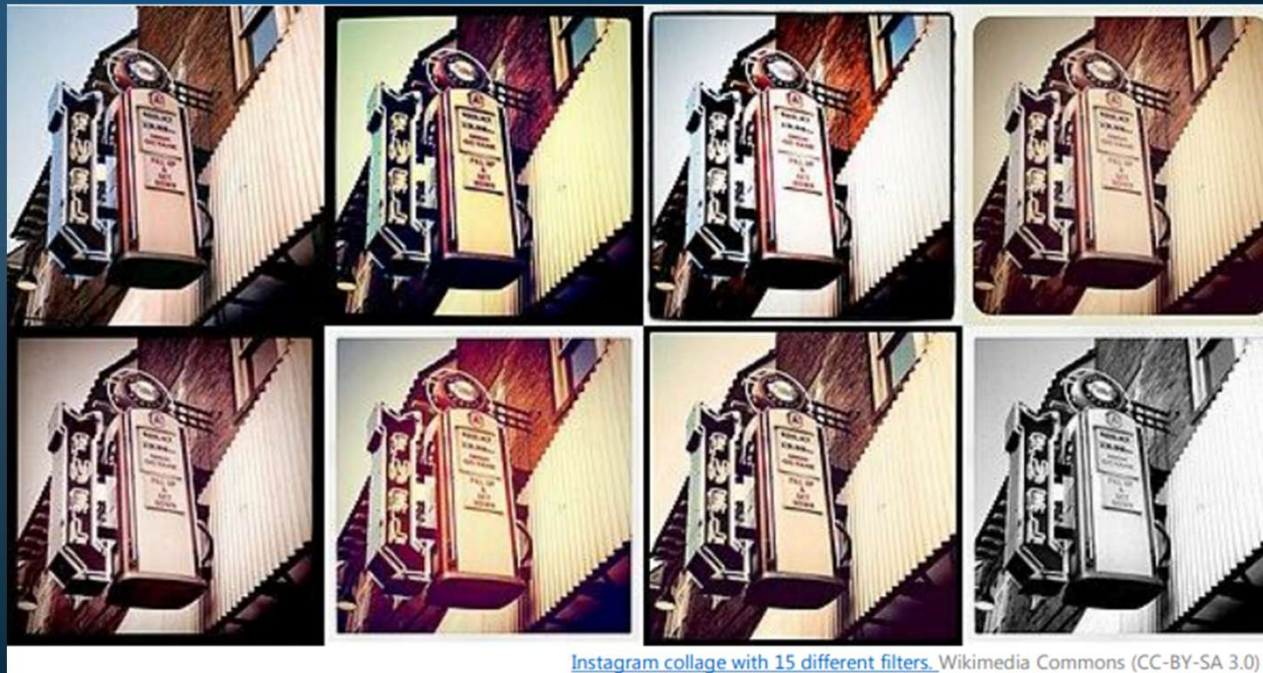


Image Filters



[Instagram collage with 15 different filters](#), Wikimedia Commons (CC-BY-SA 3.0)

Linear Filters, Denoising, Feature Enhancement, Background Subtraction

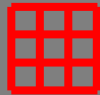
Denoising Filters: Smooth Filter

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

3x3 pixel kernel

Uniform Region

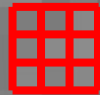
$$F \left(\begin{bmatrix} 50 & 50 & 50 \\ 50 & 50 & 50 \\ 50 & 50 & 50 \end{bmatrix} \right) = 50$$



Constant intensity regions: Output = Input

Noisy Region

$$F \left(\begin{bmatrix} 52 & 43 & 33 \\ 60 & 72 & 48 \\ 68 & 39 & 45 \end{bmatrix} \right) = 51$$



Uniform intensity with noise: Reduces intensity fluctuations

Transition Region

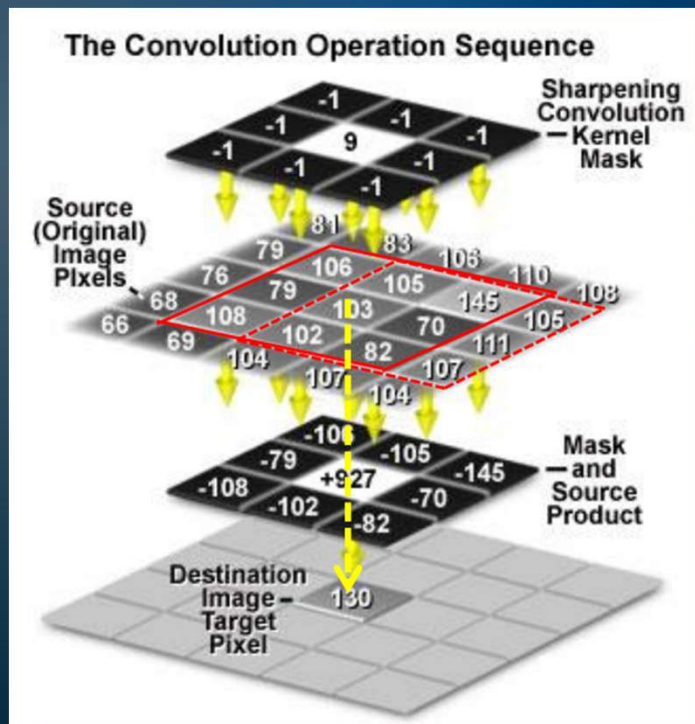
$$F \left(\begin{bmatrix} 0 & 50 & 50 \\ 0 & 50 & 50 \\ 0 & 50 & 50 \end{bmatrix} \right) = 34$$



Sharp edges: Blurred intensity transitions

The noisy intensity fluctuations are averaged out from surrounding pixels

Linear Filter (Image Convolution)



Result in Central Pixel = Weighted sum within the filter window

Weights: Filters (a.k.a. kernels, convolution matrix)

Filter is applied across the whole image to get the resulting filtered image

Image Padding

What about the edge of the image?

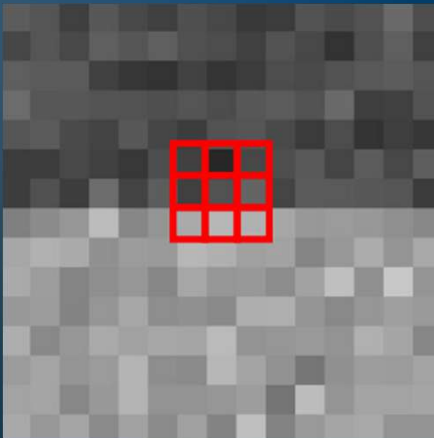
1	2	3	4	5	
6	7	8	9	1	
2	3	4	5	6	
7	8	9	1	2	
3	4	5	6	7	

Constant Padding (Zero Padding when C = 0)	Nearest/Replicate Padding	Symmetric/ Reflective Padding	Valid Padding (No Padding)																																																																																																				
<table><tr><td>C</td><td>C</td><td>C</td><td>C</td><td>C</td></tr><tr><td>C</td><td>C</td><td>C</td><td>C</td><td>C</td></tr><tr><td>3</td><td>4</td><td>5</td><td>C</td><td>C</td></tr><tr><td>8</td><td>9</td><td>1</td><td>C</td><td>C</td></tr><tr><td>4</td><td>5</td><td>6</td><td>C</td><td>C</td></tr></table>	C	C	C	C	C	C	C	C	C	C	3	4	5	C	C	8	9	1	C	C	4	5	6	C	C	<table><tr><td>3</td><td>4</td><td>5</td><td>5</td><td>5</td></tr><tr><td>3</td><td>4</td><td>5</td><td>5</td><td>5</td></tr><tr><td>3</td><td>4</td><td>5</td><td>5</td><td>5</td></tr><tr><td>8</td><td>9</td><td>1</td><td>1</td><td>1</td></tr><tr><td>4</td><td>5</td><td>6</td><td>6</td><td>6</td></tr></table>	3	4	5	5	5	3	4	5	5	5	3	4	5	5	5	8	9	1	1	1	4	5	6	6	6	<table><tr><td>8</td><td>9</td><td>1</td><td>1</td><td>9</td></tr><tr><td>3</td><td>4</td><td>5</td><td>5</td><td>1</td></tr><tr><td>3</td><td>4</td><td>5</td><td>5</td><td>4</td></tr><tr><td>8</td><td>9</td><td>1</td><td>1</td><td>9</td></tr><tr><td>4</td><td>5</td><td>6</td><td>6</td><td>5</td></tr></table>	8	9	1	1	9	3	4	5	5	1	3	4	5	5	4	8	9	1	1	9	4	5	6	6	5	<table><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>																									
C	C	C	C	C																																																																																																			
C	C	C	C	C																																																																																																			
3	4	5	C	C																																																																																																			
8	9	1	C	C																																																																																																			
4	5	6	C	C																																																																																																			
3	4	5	5	5																																																																																																			
3	4	5	5	5																																																																																																			
3	4	5	5	5																																																																																																			
8	9	1	1	1																																																																																																			
4	5	6	6	6																																																																																																			
8	9	1	1	9																																																																																																			
3	4	5	5	1																																																																																																			
3	4	5	5	4																																																																																																			
8	9	1	1	9																																																																																																			
4	5	6	6	5																																																																																																			

Not usually something to worry about in ImageJ/Fiji, but if you are writing your own code, you may have the option to decide how to handle the borders.

Denoising Filter: Median Filter

Process > Filters > Median...



$$M \left(\begin{bmatrix} 65 & 39 & 72 \\ 64 & 82 & 101 \\ 170 & 183 & 177 \end{bmatrix} \right) = 82$$

Median

[39 64 65 72 82 101 170 177 183]

Dark region pixels

Example of a **non-linear filter** → Not a weighted sum

Denoising Filter: Median Filter

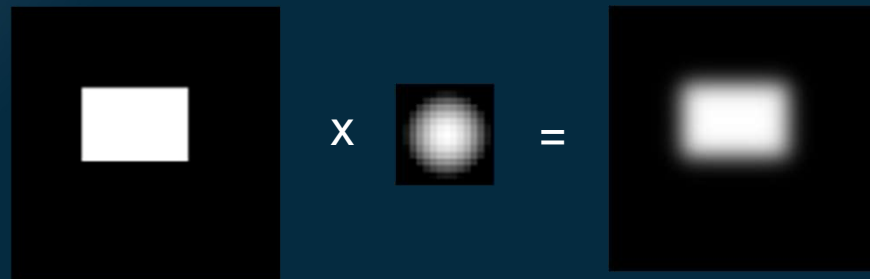
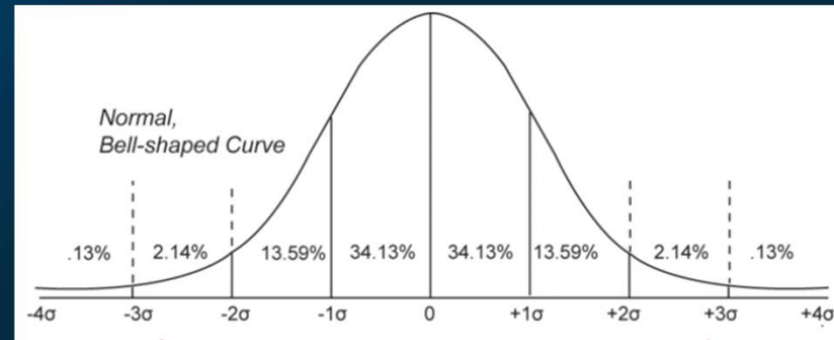
Process > Filters > Median...



The median filter attenuates noise while preserving sharp edge transitions
→ Great for removing hot pixels (salt-and-pepper noise)

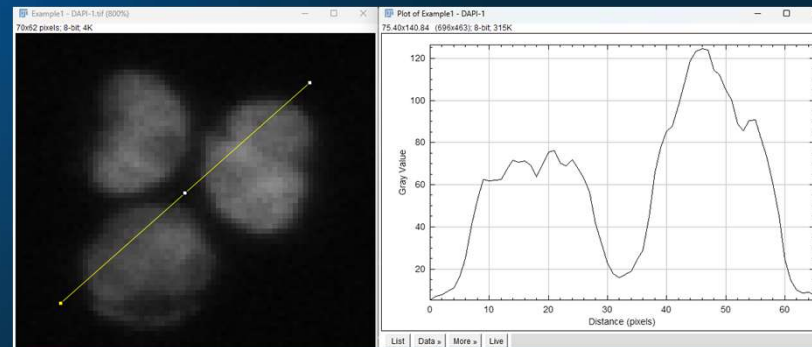
Denoising Filter: Gaussian Blur

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\left(\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

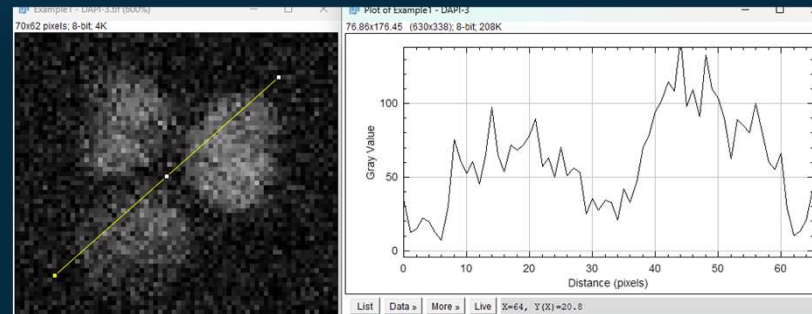


Denoising Filter: Gaussian Blur

Ideal image

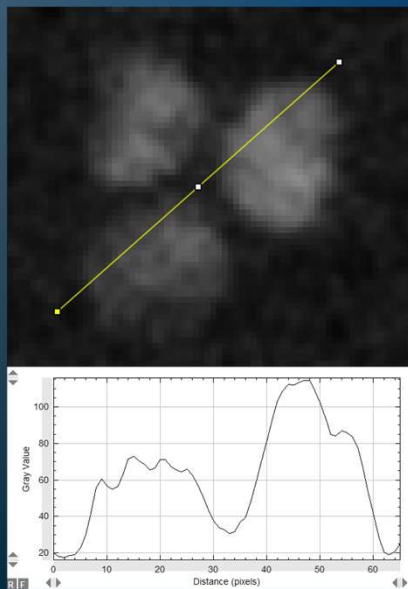


Low intensity image
(low exposure, low
laser)

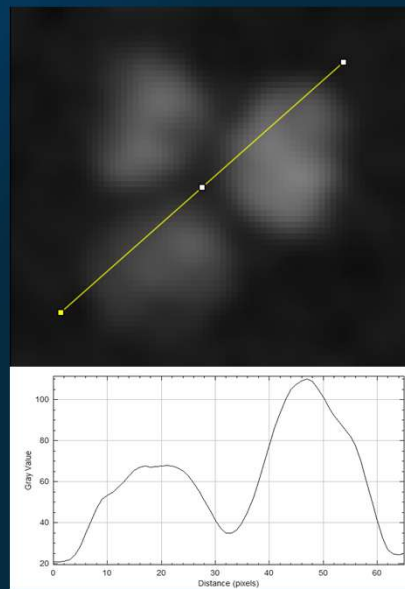


Denoising Filter: Gaussian Blur

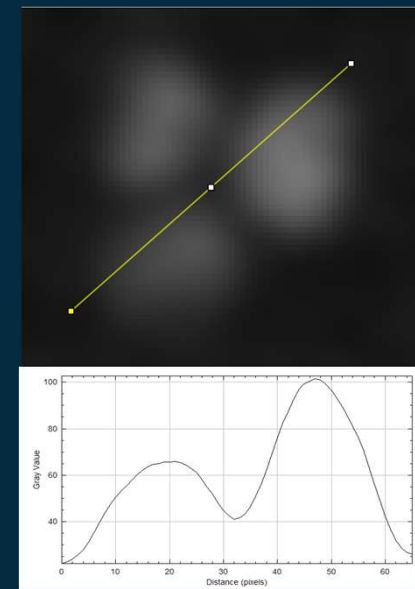
Gaussian Filter
Radius = 1 pxl

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$


Gaussian Filter
Radius = 2 pxl

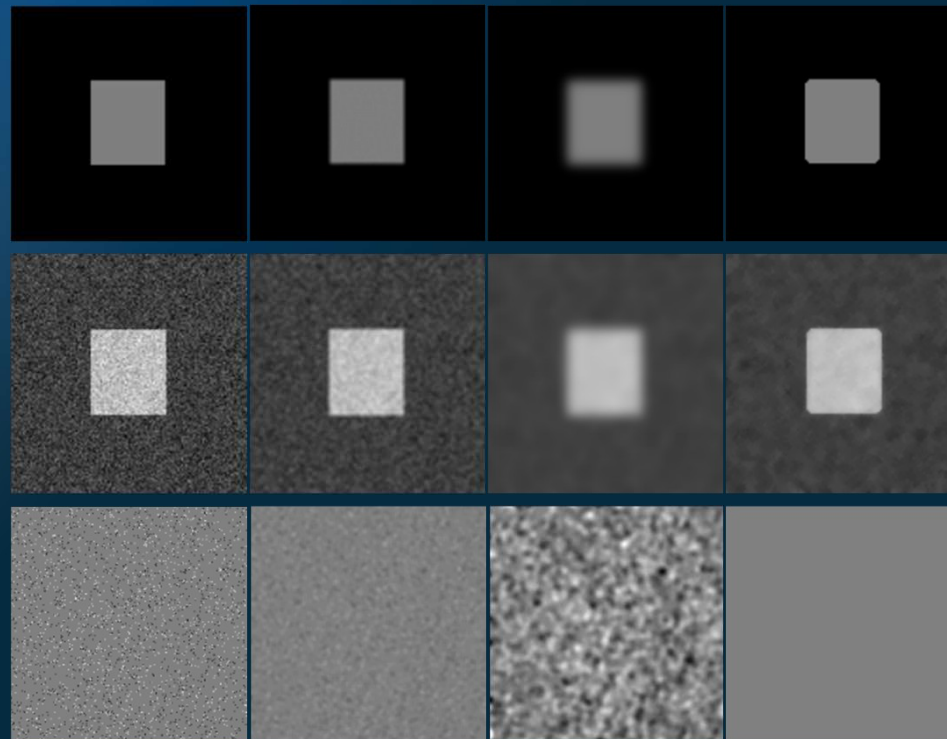
$$\frac{1}{273} \begin{bmatrix} 1 & 4 & 7 & 4 & 1 \\ 4 & 16 & 26 & 16 & 4 \\ 7 & 26 & 41 & 26 & 7 \\ 4 & 16 & 26 & 16 & 4 \\ 1 & 4 & 7 & 4 & 1 \end{bmatrix}$$


Gaussian Filter
Radius = 3 pxl

$$\frac{1}{1003} \begin{bmatrix} 0 & 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 3 & 13 & 22 & 13 & 3 & 0 \\ 1 & 13 & 59 & 97 & 59 & 13 & 1 \\ 2 & 22 & 97 & 159 & 97 & 22 & 2 \\ 1 & 13 & 59 & 97 & 59 & 13 & 1 \\ 0 & 3 & 13 & 22 & 13 & 3 & 0 \\ 0 & 0 & 1 & 2 & 1 & 0 & 0 \end{bmatrix}$$


Disclaimer: Illustration purpose for the kernels

Comparing Denoising Filters



Exercise 1. Filters and spatial resolution

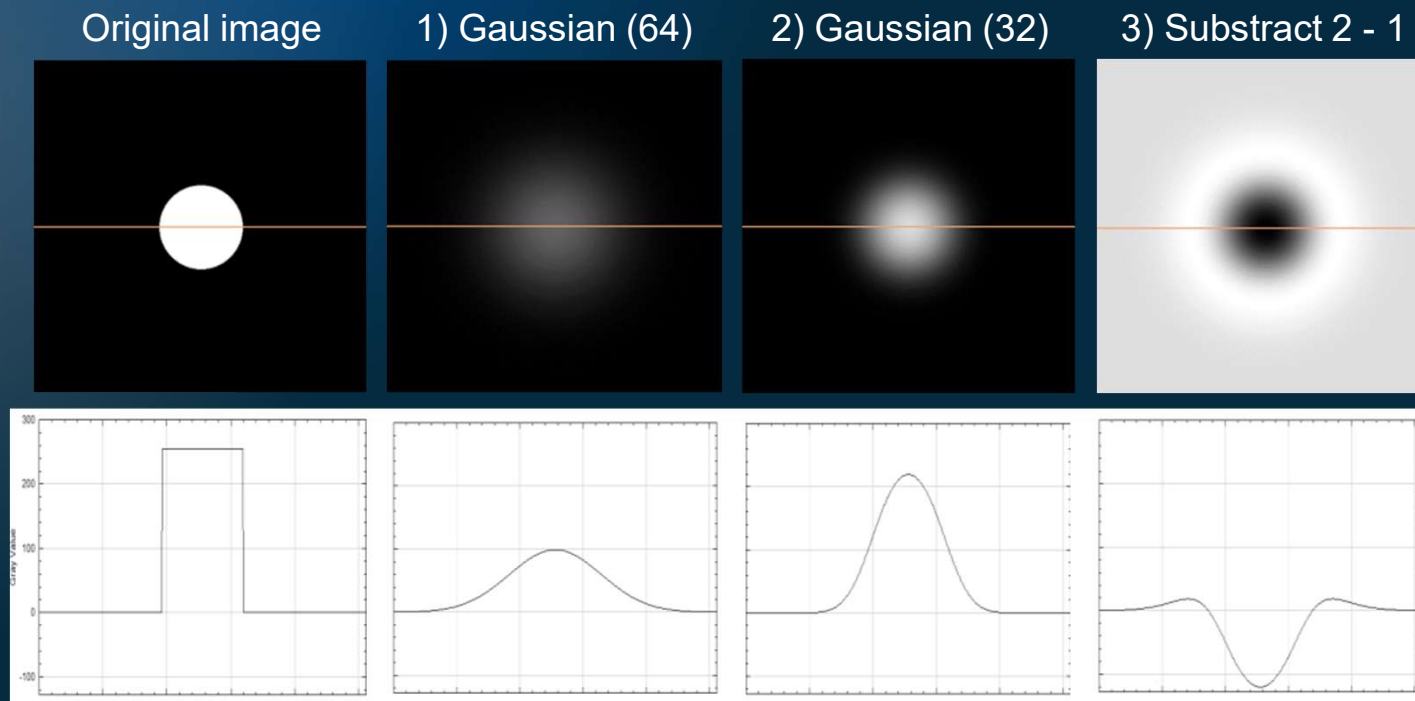
In **Process > Binary > Options...** ensure that Black Background is checked

1. Open Image Example 1
2. Filter the image with **Process > Binary > Gaussian Blur...** You can tick Preview to adjust the strength of the filter interactively: test a Radius of 0.5, 2 and 9 pixels.
3. Set the *Radius* to 2 and press OK to apply the filter.
4. Open the Threshold window (**Image > Adjust > Threshold...**)
5. Set the Threshold upper bound to 255 and optimize the lower bound
6. Classify the pixels in the image by clicking *Apply*

What problems do you observe?

Feature Enhancement: Difference of Gaussian

We can combine filters with image calculations



Feature Enhancements: Laplacian of Gaussian

Multiple filters can be applied together

Plugins > FeatureJ > FeatureJ Laplacian

Requires: ImageScience plugin

1) Gaussian Blur Kernel (σ)

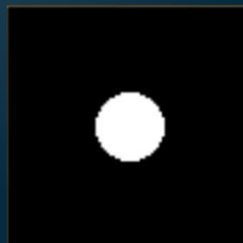
2) Laplacian Filter $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$



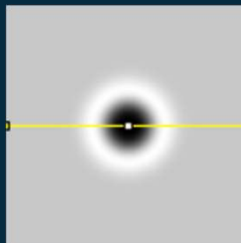
Laplacian of Gaussian Filter

0	0	3	2	2	2	3	0	0
0	2	3	5	5	5	3	2	0
3	3	5	3	0	3	5	3	3
2	5	3	-12	-23	-12	3	5	2
2	5	0	-23	-40	-23	0	5	2
2	5	3	-12	-23	-12	3	5	2
3	3	5	3	0	3	5	3	3
0	2	3	5	5	5	3	2	0
0	0	3	2	2	2	3	0	0

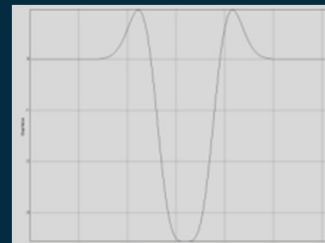
Discrete approximation
of LoG kernel for $\sigma=1.4$



Original Image



LoG Filtered

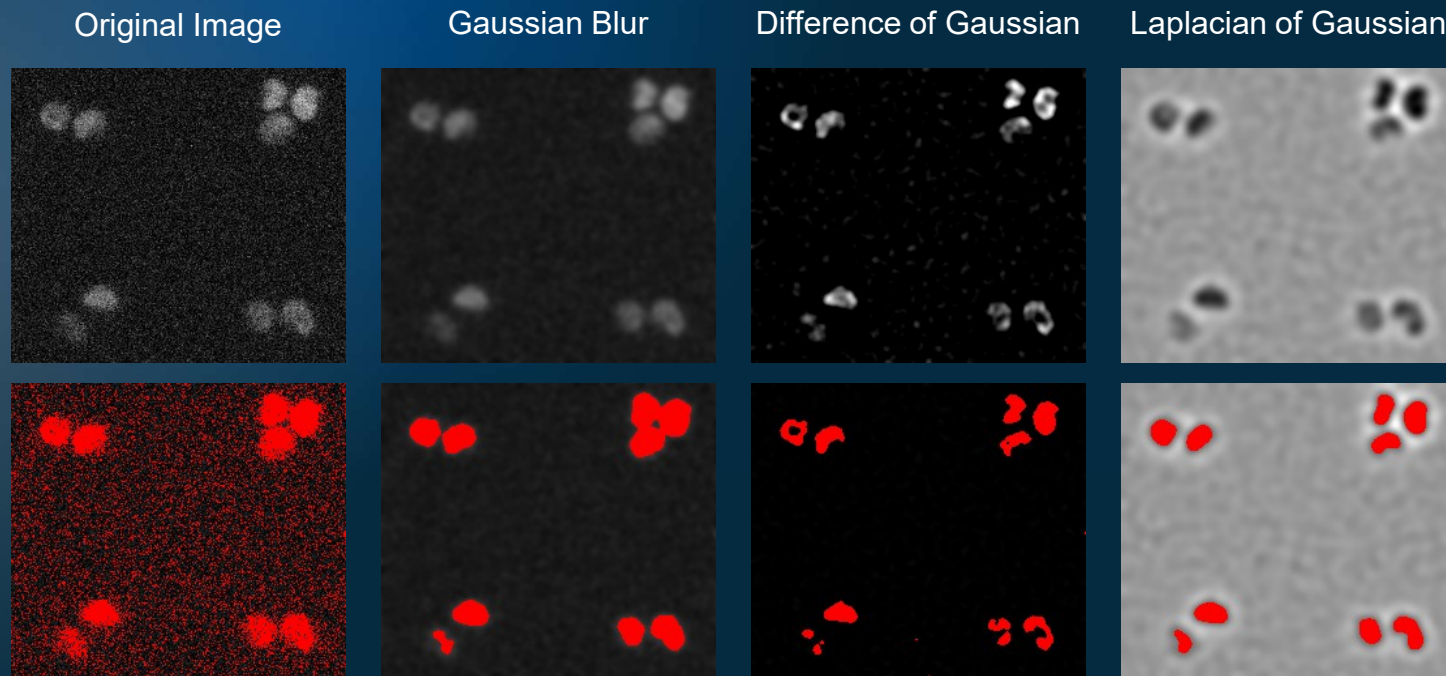


Intensity Profile

Maximum contrast

$$\sigma = \frac{r}{2}$$

Feature Enhancements: blobs (Ellipsoids)



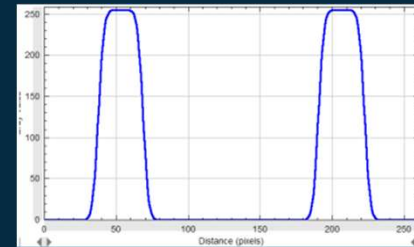
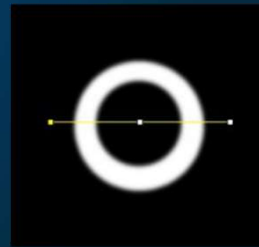
Object Model: Blob-like (ellipsoids) of a characteristic size
Blob shapes highlighted by DoG and LoG filters makes thresholding easier

Exercise 2.1: Blob enhancement

1. Open Image Example 2
2. Perform the same operations as in Exercise 1 (filtering and thresholding) but this time apply the Laplacian of Gaussian instead of the Gaussian filter (Plugins > FeatureJ > FeatureJ Laplacian)
3. Optimise the Smoothing Scale (Radius) to obtain the best results. The LoG filter cannot be previewed – estimate the characteristic size of the object and use it to estimate the theoretical optimal Smoothing Scale.
 - When testing parameters without preview you can use Undo (Ctrl+Z) – only works on processing 2D images.
 - ImageJ only allows one undo step. The better practice is to keep a copy of the original image by duplicating it (Image > Duplicate or Ctrl+Shift+D) when testing different methods and parameters.
 - Why is the resulting image 32-bits?
 - Apart from blobs, what else is emphasized in the processed image?

Feature Enhancement: Edge Detection

Edges → Sharp changes in brightness



Calculate image gradient

Sobel edge detector (Process > Find edges)

Horizontal gradient (Gx)

$$\begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

Vertical gradient (Gy)

$$\begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

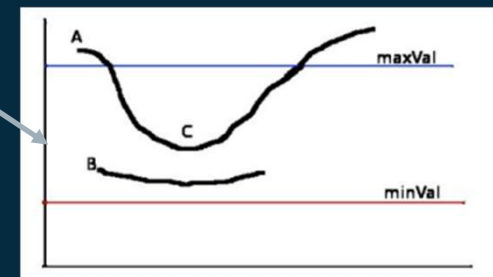
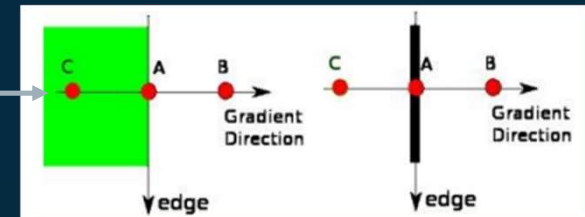
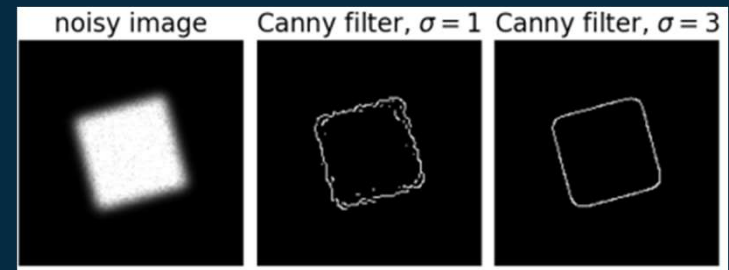
$$G = \sqrt{G_x^2 + G_y^2}$$

Feature Enhancement: Edge Detection

Multi-step algorithm: Canny Edge Detection

1. Noise reduction with Gaussian filter
2. Calculate intensity gradients and edge detection with Sobel kernels
3. Edge thinning by suppressing pixels that are not the neighbourhood maxima in the gradient direction
4. Classify edges based on input maxVal and minVal
5. Discard weak unconnected edges that are below minVal

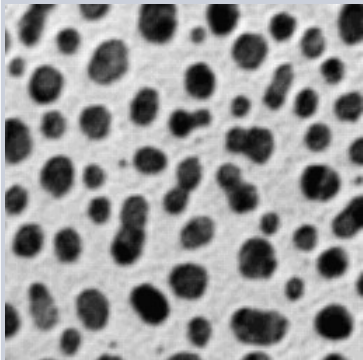
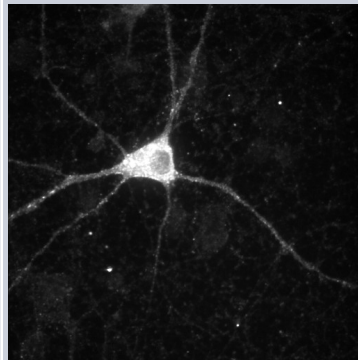
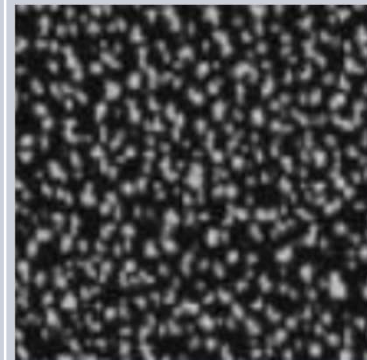
Plugins > Feature Extraction > FeatureJ > FeatureJ Edges



Exercise 2.1: Blob enhancement

1. Open Image Example 2
 2. Call **Plugins > FeatureJ > FeatureJ Edges** with:
 1. Compute gradient-magnitude image
 2. Smoothing scale = 3
 3. Suppress non-maximum gradients
 3. Manually optimise the global thresholder to segment the image.
 4. Call **Process > Binary > Fill Holes**
- What does Step 4 do? What do you think of this segmentation?
- We will learn more about working with the resulting binary mask later.

Other filters for feature enhancement

Feature	Blobs	Edges	Tubes/Vessels	Speckles
Example image				
Possible filters	LoG, Dog	Canny, Prewitt, Roberts, Sobel, LoG, DoG	Meijering, Sato, Frangi, Hessian	Top-hat

Background Subtraction



Process > Filters > Minimum... (30)

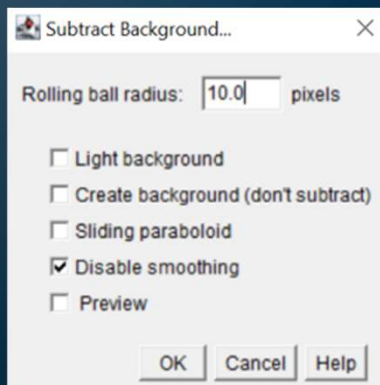
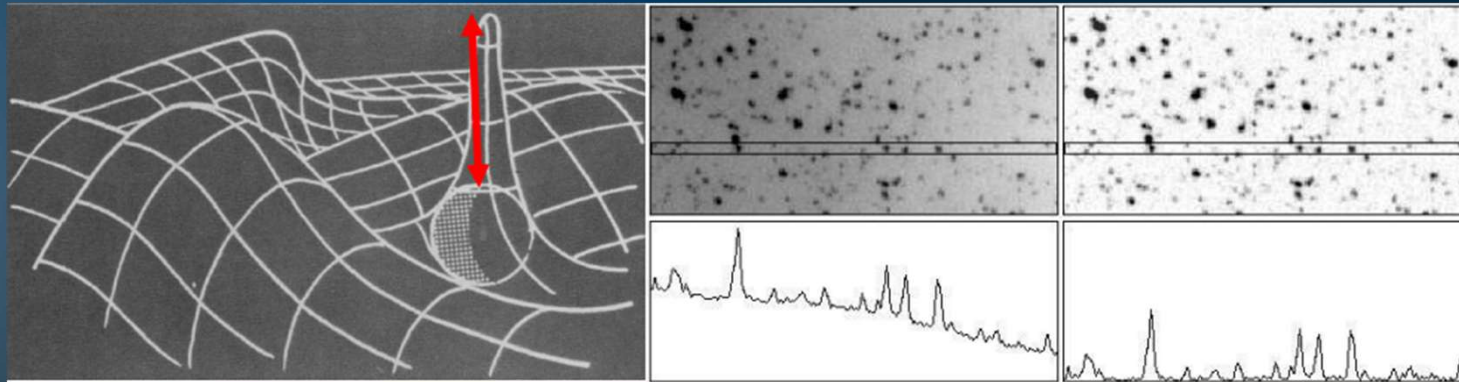
Process > Filters > Maximum... (30)

Process > Image Calculator...

Image 1: Original

Image 2: Max of Min

Background Subtraction: Rolling-ball Algorithm



For each pixel, the grayscale level reached by the tip of the ball is **Subtracted** to the image

Process > Subtract Background

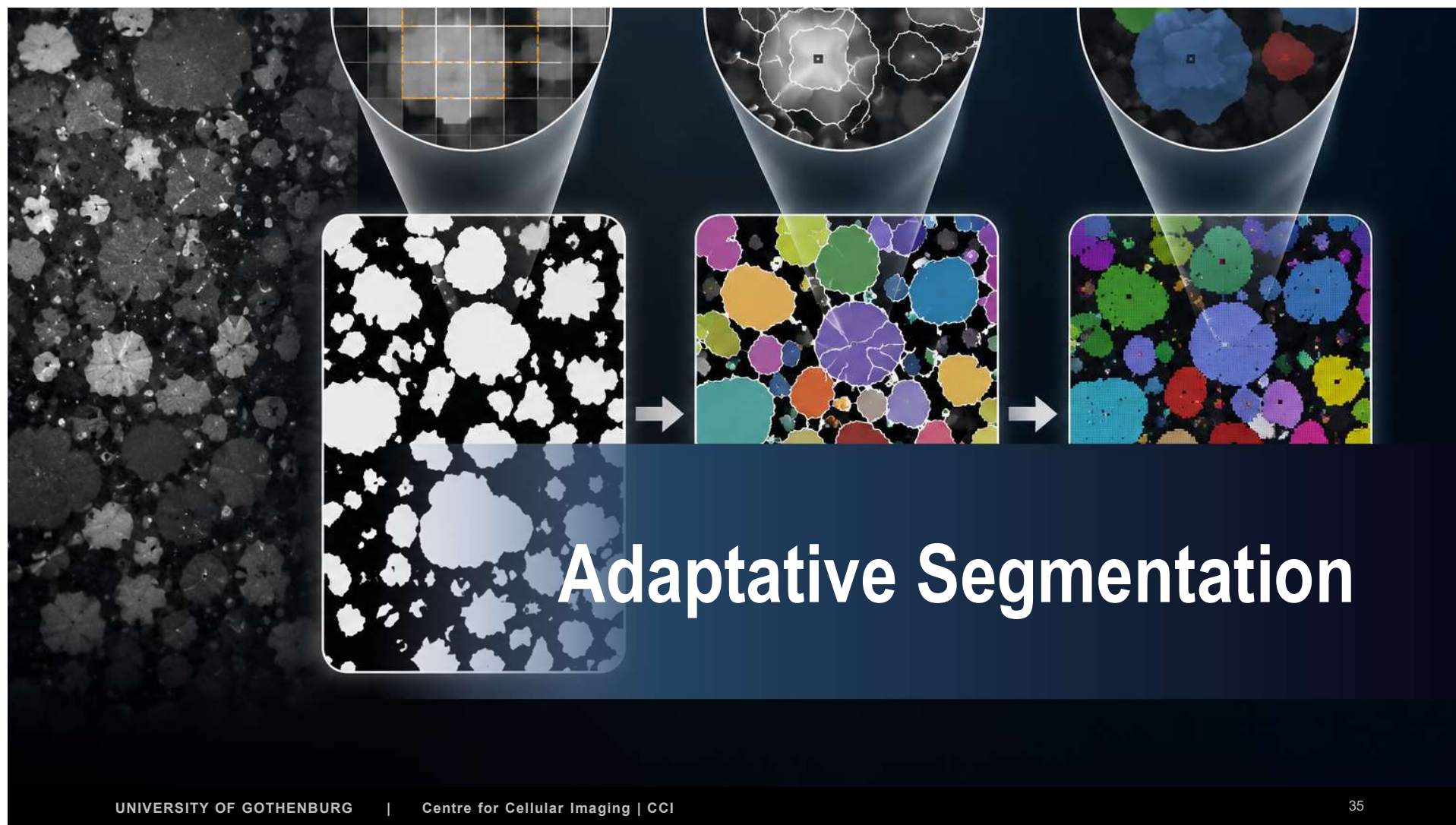
- Light background: Dark object/bright background
- Create background: Returns background
- Sliding paraboloid: Approximation (Faster)
- Disable smoothing: Ensure results > 0

Exercise 3: Background Subtraction

1. Open Image Example 3.
2. Use the line tool to estimate the size of a speckle.
3. Call Process > Filters > Top hat...
4. Tick *Preview* and see what happens when the radius input is much smaller or larger than the speckle size.
5. Select OK when you are happy with the results
6. Try steps 3-4 again, this time with Subtract Background, and compare the outcome.

Key Concepts So Far

- Digital images as mathematical objects
- Image processing can be done to create derivative images with unwanted features removed or features of interests highlighted. This helps improve segmentation for objects of interest.
- Object models allow us to decide what filters to apply at what size
- Many different algorithms exist to do similar processing steps.
- Image processing steps can be of varying complexity.



Heterogeneous Intensity Based Thresholding

Original Image



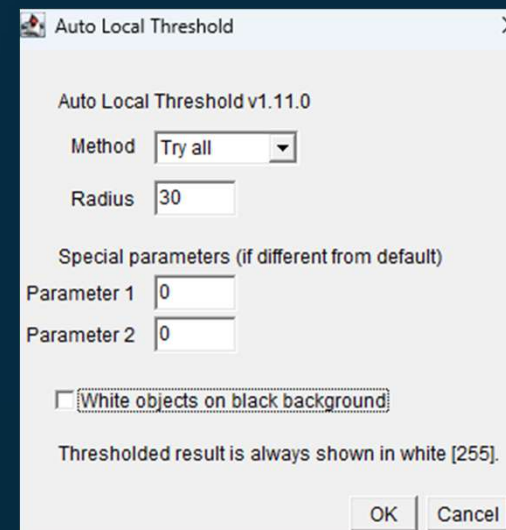
Otsu's Thresholding



For very heterogeneous object intensities, global thresholding is not well adapted because the optimal threshold is **different** for each object.

Local Thresholding: Mean Method

Image > Adjust > Auto Local Threshold



1. For all possible pixels, compute Mean Intensity, M , inside a disk of a given Radius
2. Pixels with intensity $\geq M + \text{Offset}$ (Parameter 1) are classified as foreground

Assumption: Local mean Radius $\gg$ largest object

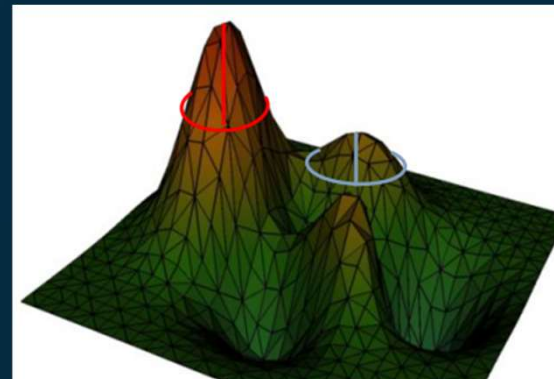
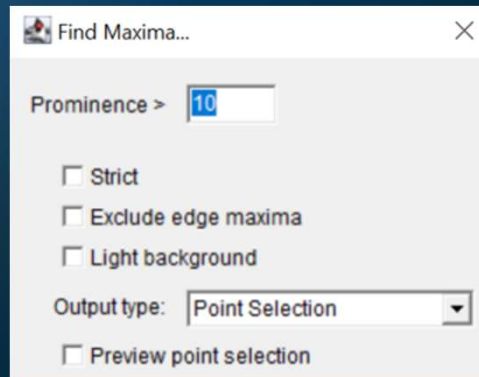
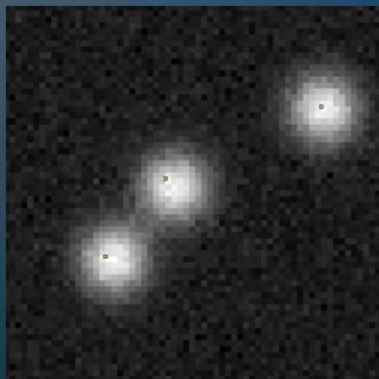
Exercise 2.3: Local thresholding

1. Open Image Example 2.
 2. Apply median filtering (Process > Filter > Median...) with Radius = 2
 3. Perform Local Thresholding (Image > Adjust > Auto Local Threshold) with Mean Method, Radius = 25, and Offset (Param 1) = -4
- What do you observe? Is the segmentation of the object more accurate than before? Are there any errors and what is their cause?
 - What happens when an object of interest is significantly smaller than the radius of the median filter

Local and Regional Maxima (2D)

Local Maximum: Pixel that is greater than or equal to all its neighbours

Process > Find Maxima...

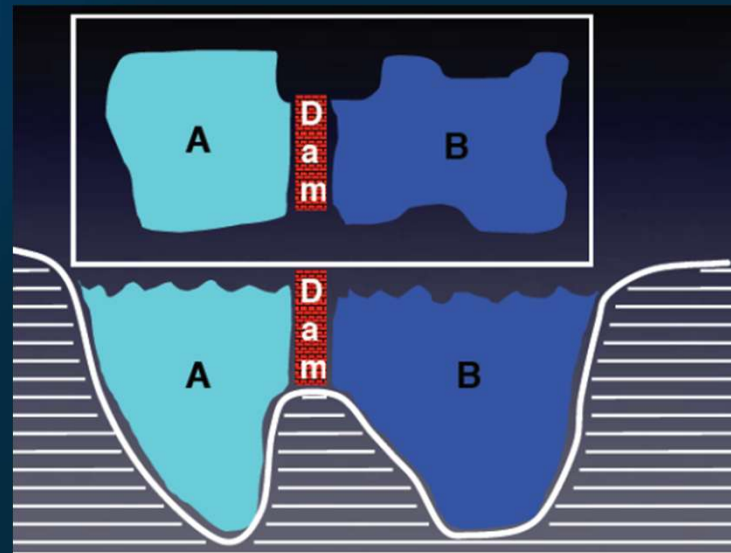
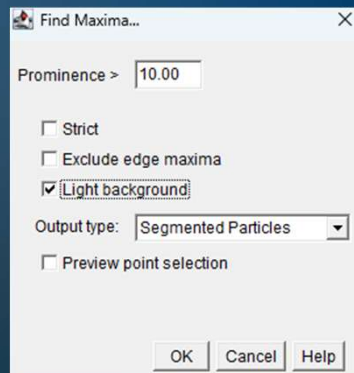


By adjusting Prominence (**red line**), the weak **blue** maximum is discarded

Regional Maxima is more adaptive to spatial context

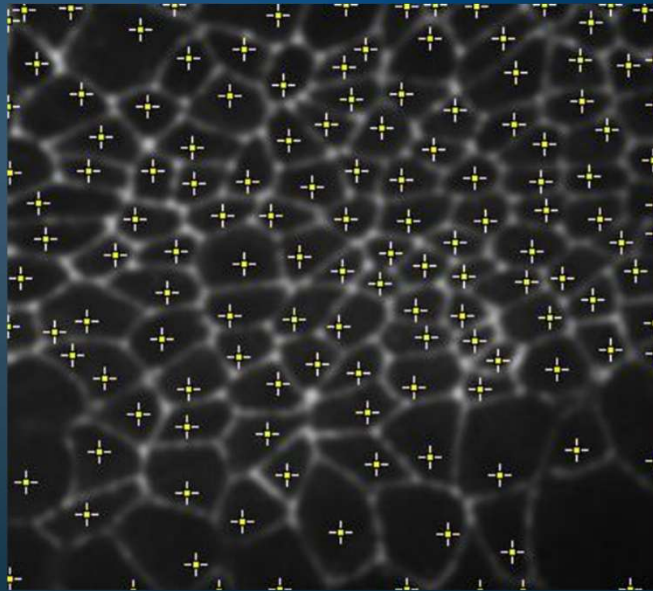
Seeded Watershed Segmentation

Process > Find Maxima...

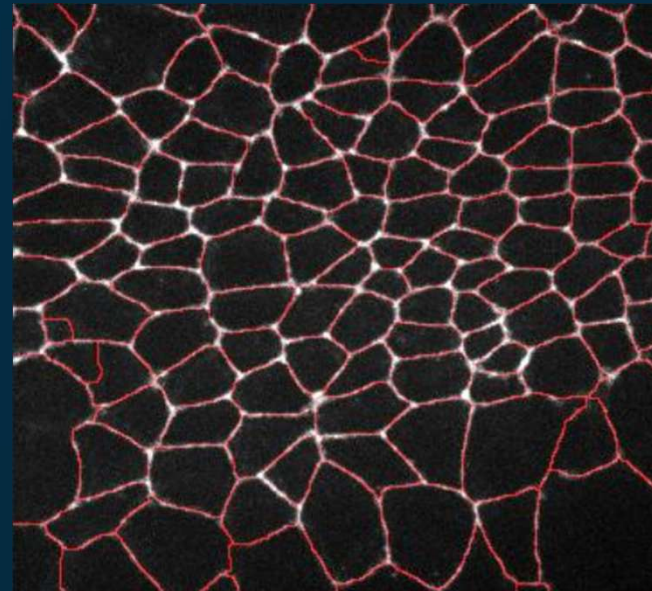


1. Compute Regional Minima (Seeds)
2. Flood basins from minima
3. Draw Watershed Lines where two basins merge

Seeded Watershed Segmentation



Gaussian Blur (Rad = 2 pix)
+ detect Regional Minima
(Prominence = 3)



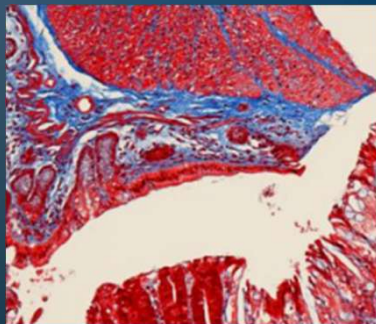
Seeded Watershed from
Regional Minima

Exercise 4: Watershed Segmentation

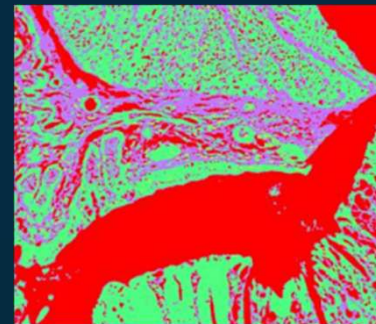
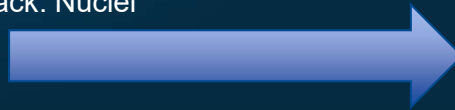
1. Open Image Example 4
 2. Apply **Process > Filter > Gaussian Blur...** with Rad = 1.5 px
 3. Call **Process > Find Maxima...** with the following settings:
 1. Light Background – Finding intensity minima
 2. Preview point selection – Use this to adjust *Prominence* to have one marker in most objects (around 3 gray levels).
 3. Output Type: *Segmented Particles*
 4. Select mask image and call **Edit > Invert** (Ctrl+Shift+I)
 5. Call **Image > Color > Merge Channels...** and assign the filtered image to the grey channel and the mask image to the red channel.
 1. Create composite
- You can use **Image > Color > Channels Tool...** (Ctrl+Shift+Z) to toggle Channel 1 and check the segmentation quality.

Machine Learning Pixel Classification

What if you want to split your image into more than 1 class and assign EVERY pixel into a class?



White: Background
Red: Muscle
Blue: Collagen
Black: Nuclei

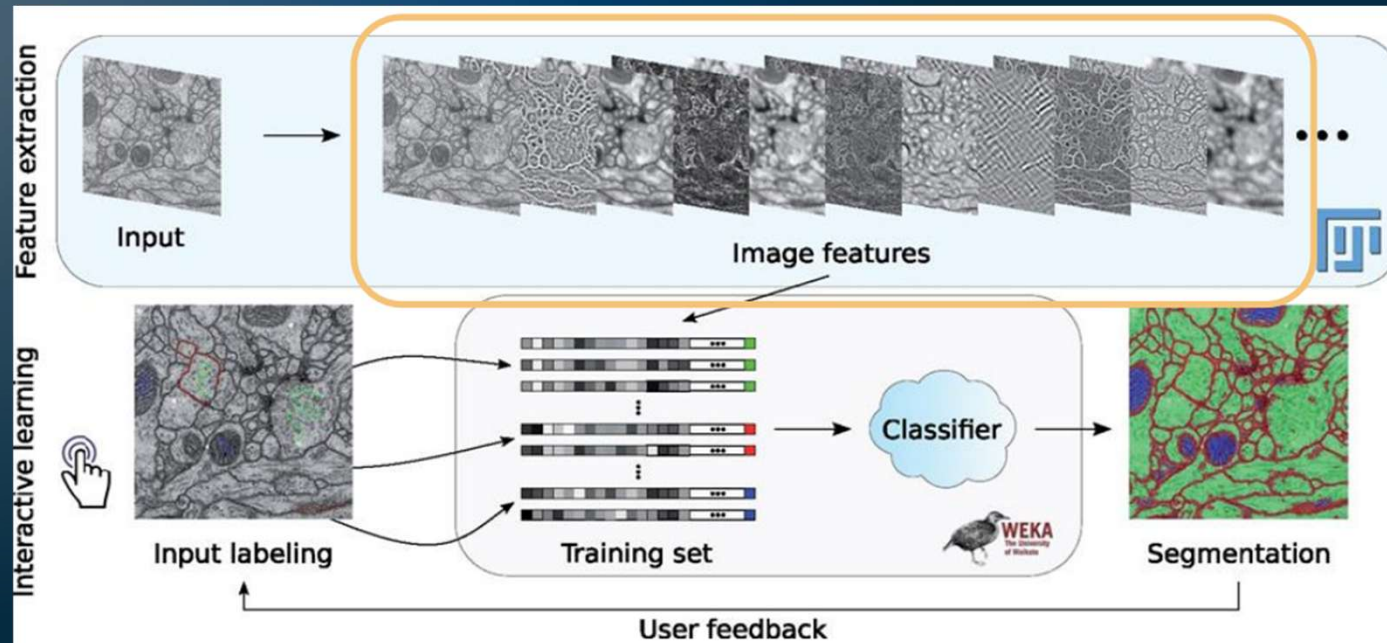


Red: Background
Green: Muscle
Purple: Collagen
Yellow: Nuclei

What might happen if you segment each signal one-by-one?
→ Conflicting class assignments

Machine Learning Pixel Classification

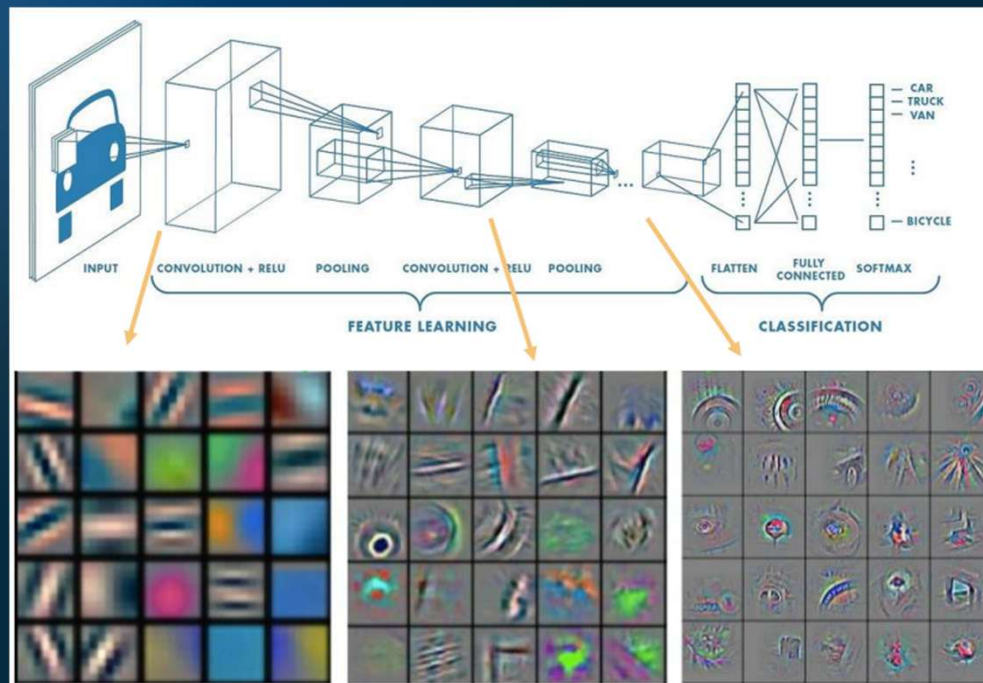
Demo with Plugins > Segmentation > Trainable Weka Segmentation



Multiple features applied on each channel at different scales → rich context!

Machine Learning Pixel Classification

Convolutional neural network (deep learning) learns image features too.

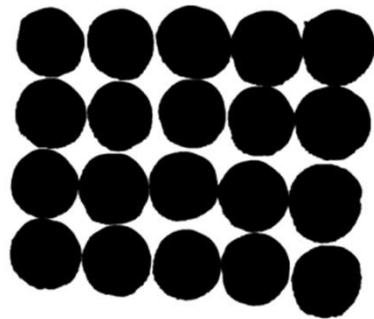


Key Concepts So Far

- 1. Having spatial context is very important for identifying regions of interests in images
- 2. Spatial filters allow us to account for the local context of each pixel.
- 3. Adaptive methods allow us to account for greater amounts of spatial context



General Image Analysis Workflow



Number of
tomatoes: 20
Average tomato
diameter: 6cm

Image

Segmentation

Objects

Measurements

- How do we get separate objects?
- What if our masks are still untidy, can we do anything to clean it up?

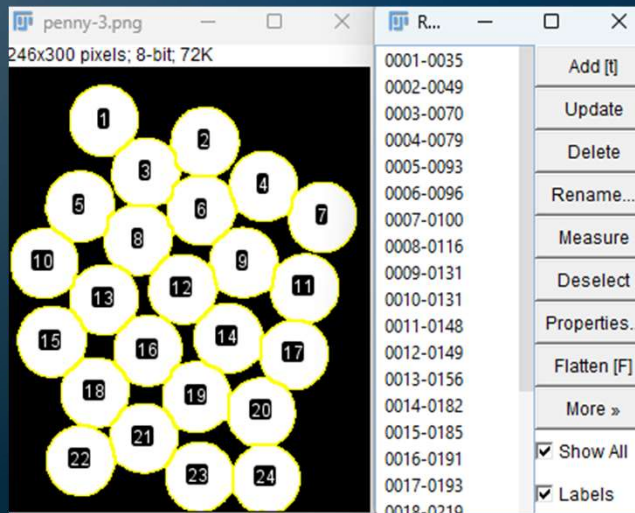


Representing Segmented Objects

Polygons (Regions of Interest)

Each object is represented by a set of coordinates that outlines it.

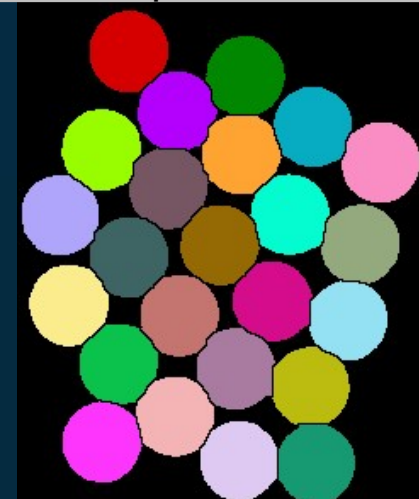
Analyze > Analyze Particles



Label Mask

Each object (connected particle) is represented by giving its mask a different value.

Plugins > MorphoLibJ > Binary Images > Connected Component Labelling

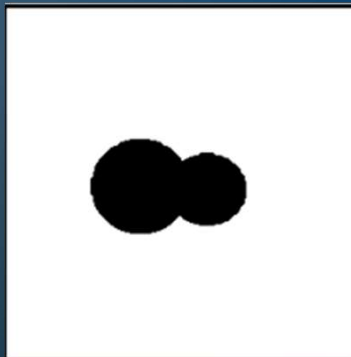


Binary Watershed for Splitting Particles

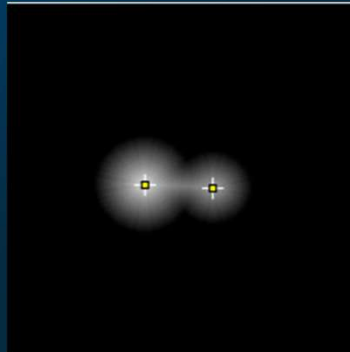
Process > Binary > Watershed

OR

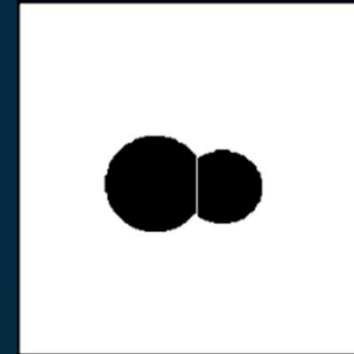
Plugins > MorphoLibJ > Binary Images > Distance Transform Watershed



Original Mask



Distance Transform



After Binary Watershed

1. Distance Transform: Get minimum distance to background
2. Seeded Watershed from Intensity Maxima

Assumption: Objects are approximately convex

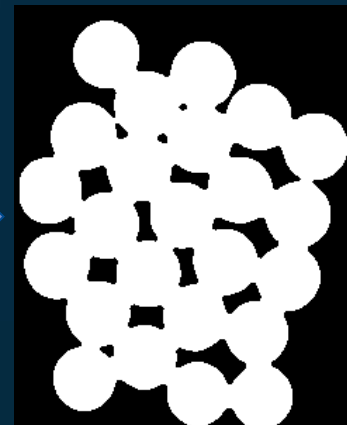
Binary Watershed: Example Workflow



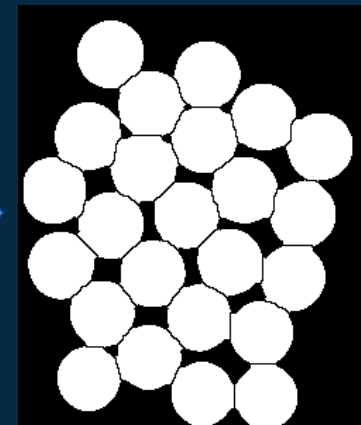
Original Image



Subtract
Background



Median Filter +
Otsu Threshold



Binary Watershed

Exercise 2.4: Distance Transform Watershed

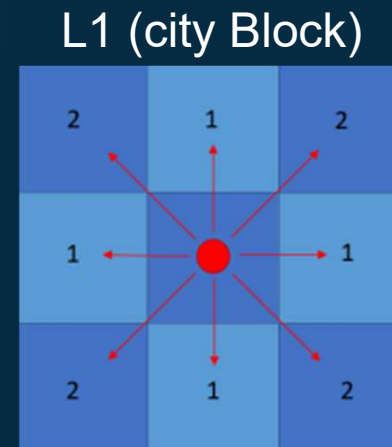
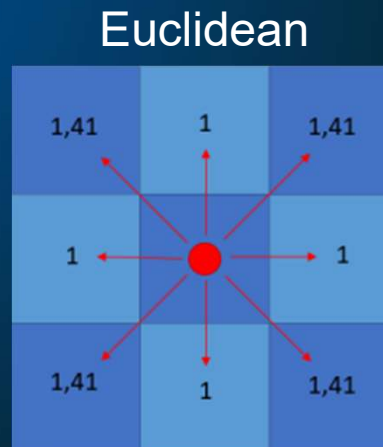
Let's try Exercise 2 one more time!

1. Open Image Example 2
2. Apply `Process > Filters > Median...` with Rad = 2 px
3. Apply a global threshold to the image (T1 = 45, T2 = 255)
4. Duplicate the resulting mask, and name one "ROI" and the other "Label"
5. For "ROI", call `Process > Binary > Watershed` and then use `Analyze > Analyze Particles` (Tick Add to Manager)
6. For "Label", Call `MorphoLibJ > Binary Images > Distance Transform Watershed` and check Preview.

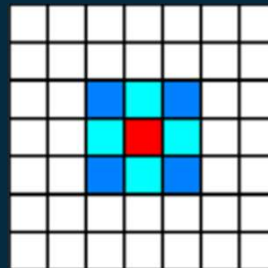
- What happens when you change Distance and Connectivity?
- The Distance Transform Watershed algorithm can be sensitive to object holes and jittery edges (may lead to spurious splits). Remember to filter the image and fill small holes prior to applying it.

Distance Metric and Connectivity

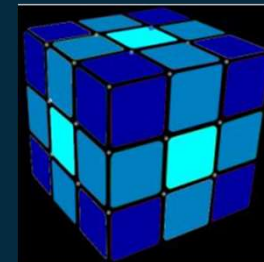
Distance metric:
Methods for
quantifying distance.



Connectivity
A connected particle
is a group of
connected pixels



Central pixel is 4-
connected to the
light blue square
and 8-connected
to both light and
dark blue square

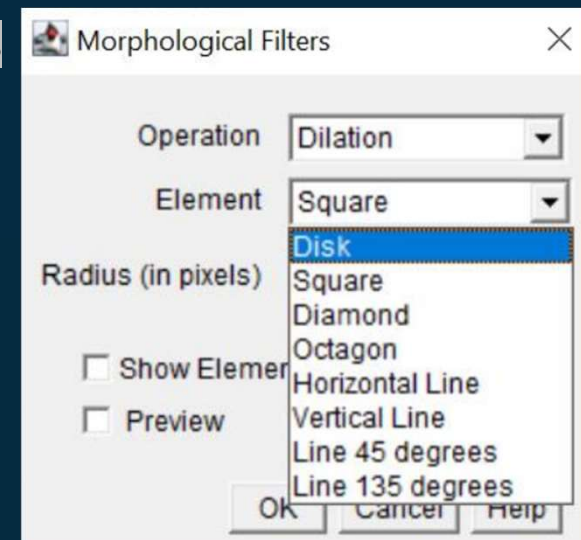
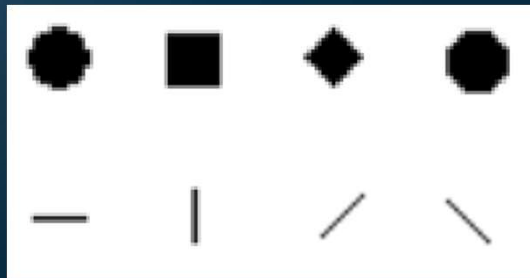


Central voxel is 6-
connected to bright
blue voxels and 26-
connected to all
neighbour voxels

Morphological Filters

A morphological filter uses a sliding window of user defined shape (structuring element) to implement nonlinear operations (e.g. min, max) inside the window.

Plugins > MorphoLibJ > Filtering > Morphological Filters

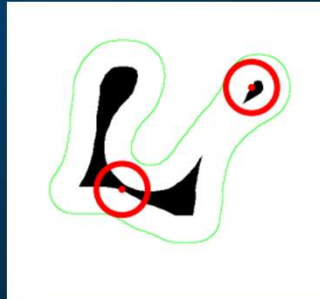


Morphological Filters: Basic Operations

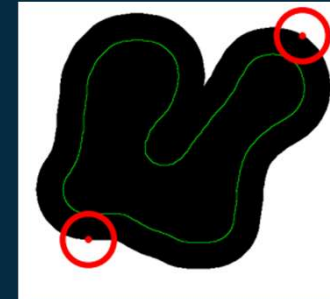
Original Mask



Erosion: Min Intensity



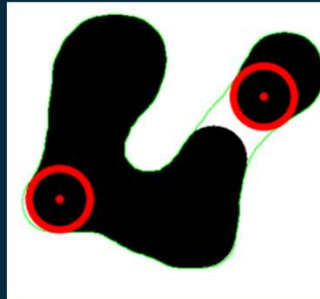
Dilation: Max. Intensity



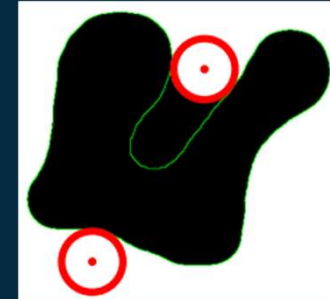
Structuring Element:
Disk



Opening: Erosion +
Dilation



Closing: Dilation +
Erosion



Morphological Filters: Discarding Small Particles

A: Original Mask



B: A Eroded by a Disk



C: B Dilated by same Disk

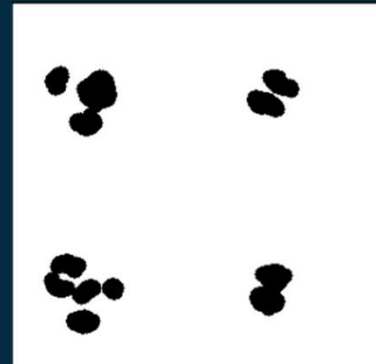
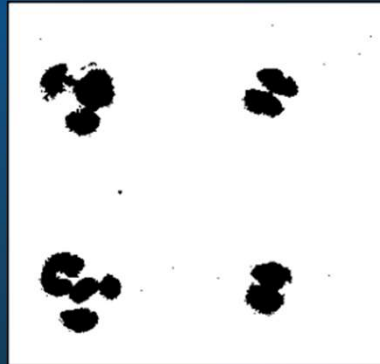


Difference Image A - C



Morphological Filters: Cleaning and Selection

Mask Cleaning
(Opening by a Disk)



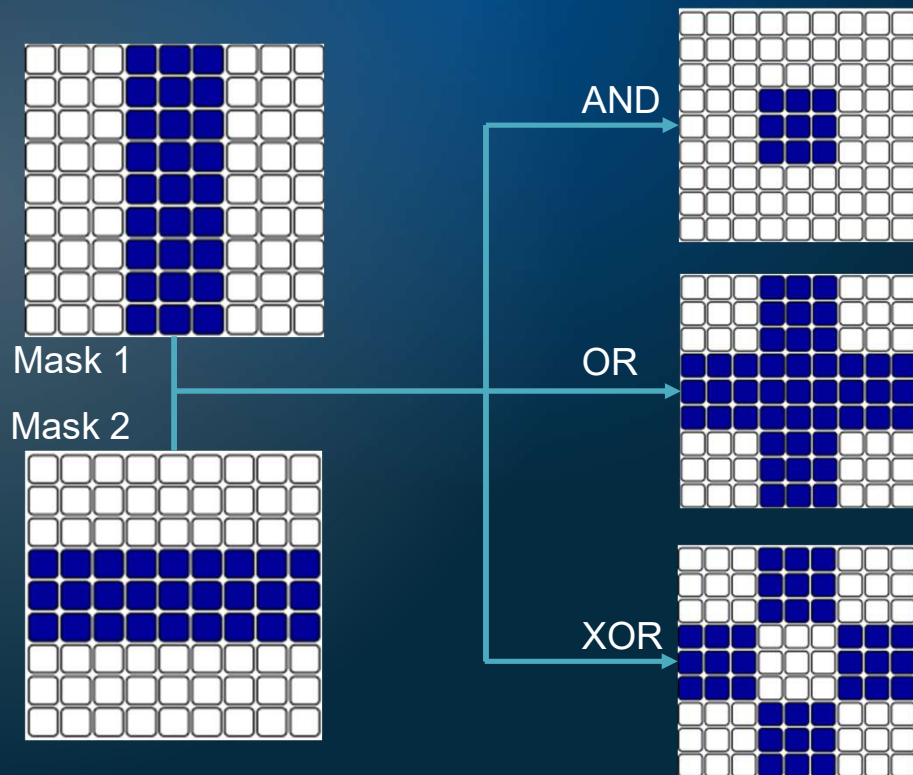
Geometry Selection
(Opening by a Vertical Line)



Original

After morphological filter

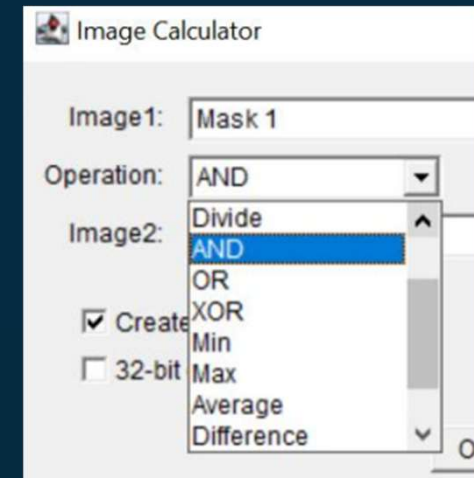
Logical (Boolean) Operations



AND	0	1
0	0	0
1	0	1

OR	0	1
0	0	1
1	1	1

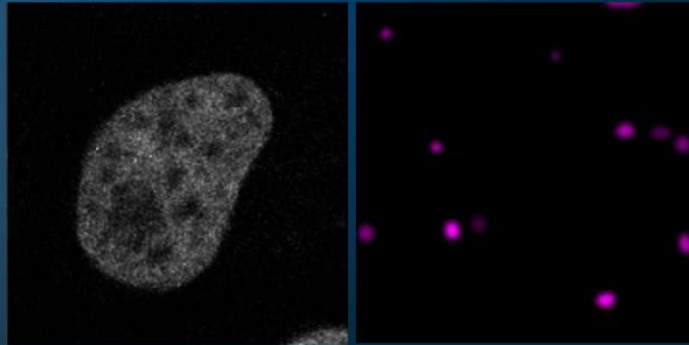
XOR	0	1
0	0	1
1	1	0



Logical (Boolean) Operations

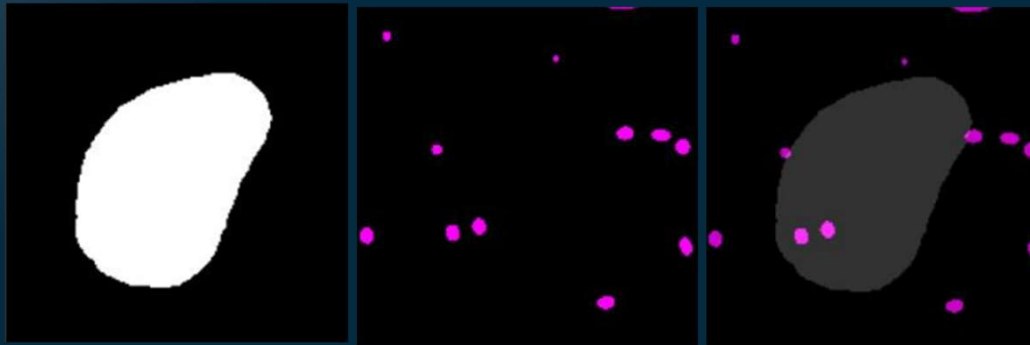
Example usage – Finding protein in nuclei

Intensity Image



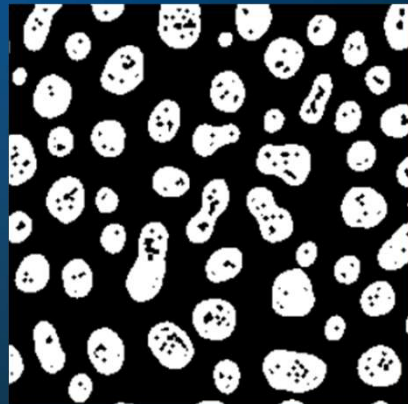
Which operation should be used?

Mask



Other Useful Binary Operations

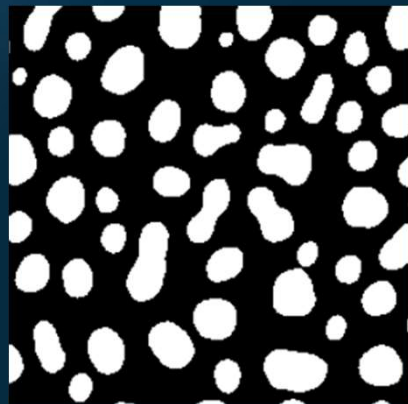
Original Mask



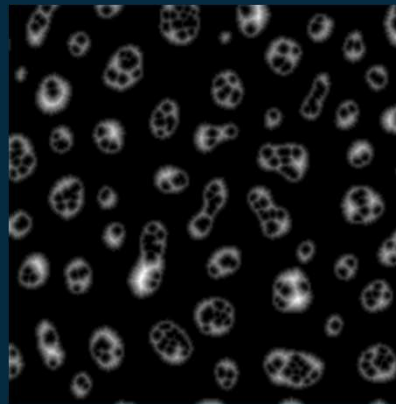
Clear Borders



Fill holes

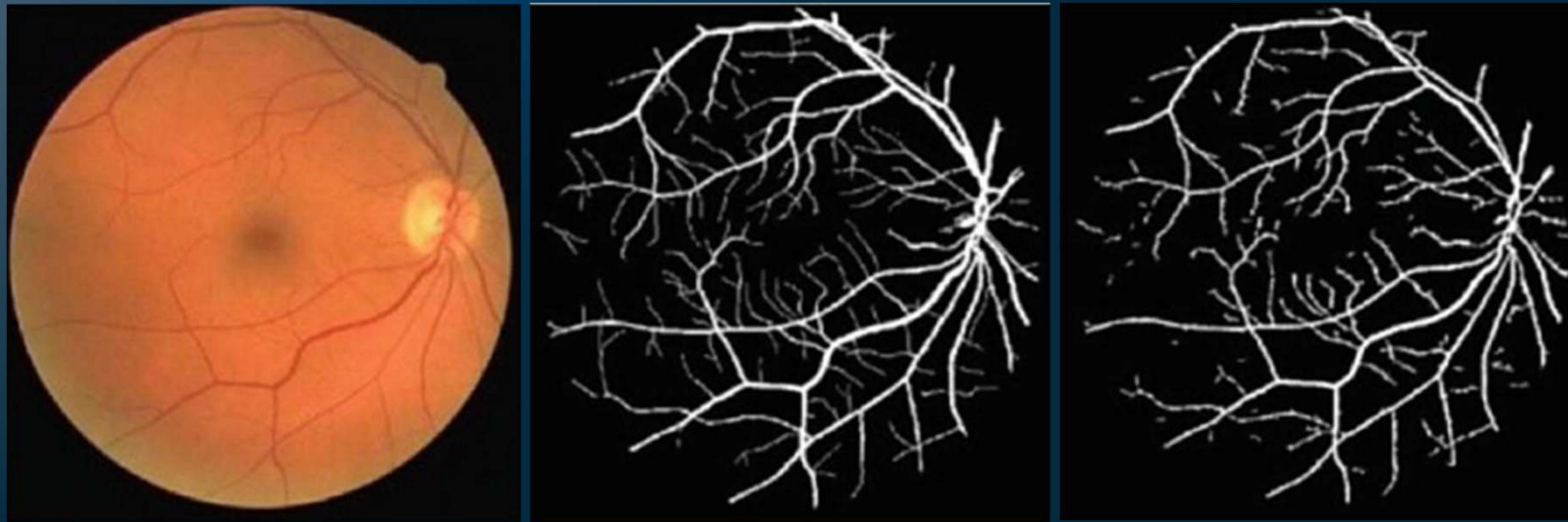


Distance Transform



Other useful binary operations: Skeletonisation

Reducing binary mask down to create a 1-pixel wide line that representation of the mask.



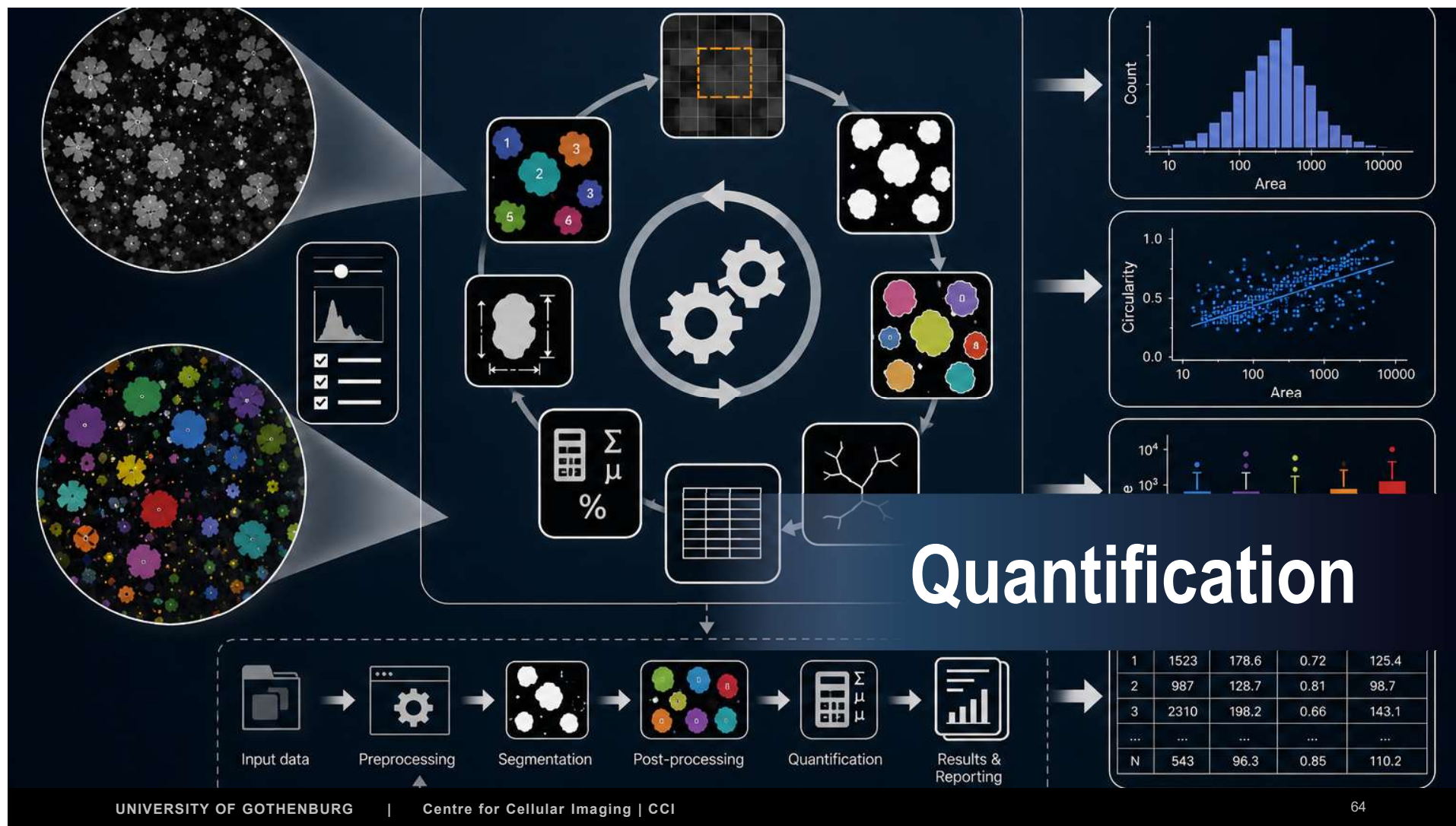
Exercise 5: Binary Mask Processing

1. Open Image Example 5
2. Call `Plugins > MorphoLibJ > Filtering > Morphological Filters` and check Preview.
3. Adjust Operations, Elements and Radius to reconnect the disconnected branches of the neuron.
4. Fill any holes in the skeleton with `Process > Binary > Fill Holes`
5. Duplicate the resulting mask
6. On one image call `Process > Binary > Skeletonize`
7. On the other image `Process > Binary > Distance Map`
8. Use the image calculator to find the results of Skeleton AND Distance Map
 1. You may have to adjust Brightness/Contrast to see the results
 2. What does the image result represent?

A note on image data handling

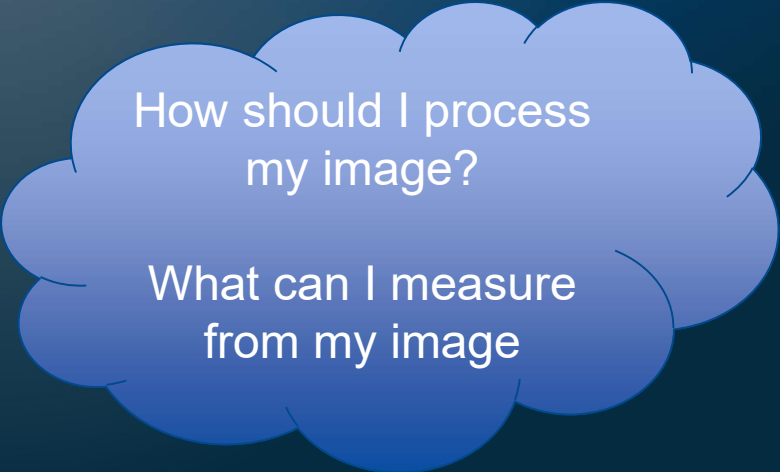
We have manipulated our image a lot from its original form through image processing... When is this okay?

- Purpose of processing?
 -  Identifying ROIs computationally (vs. manual drawing)
 -  Quantifying intensity on processed images
- Original image still available?
- Are processing steps well documented?
- Are there experimental controls?



Quantification and Analysis

When designing your image analysis workflow, it is important to keep the entire experiment objective in mind.



How should I process my image?

What can I measure from my image

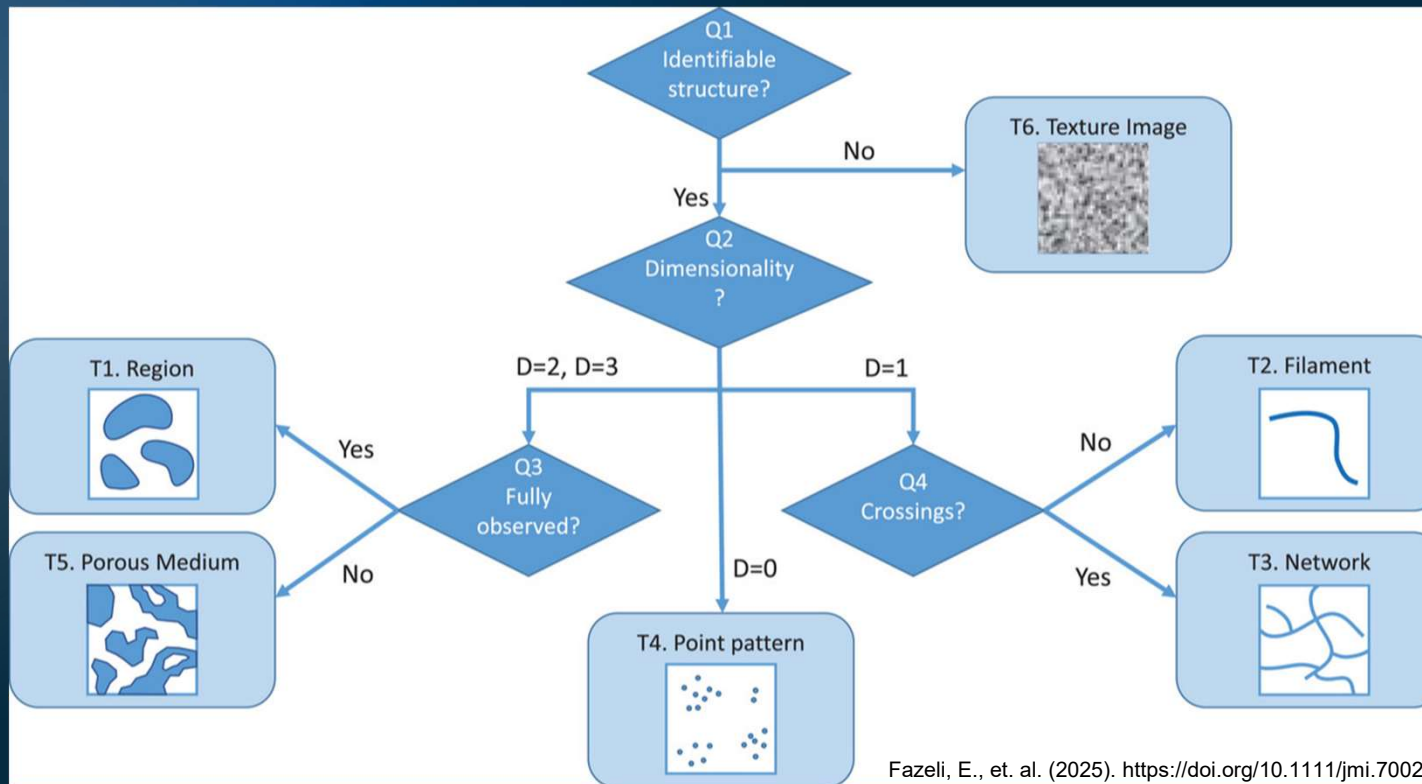
What is your scientific questions?

What metric would answer it?

Can you see and quantify relevant signal?

Do you need absolute or relative measure?

Quantification: Structure Type

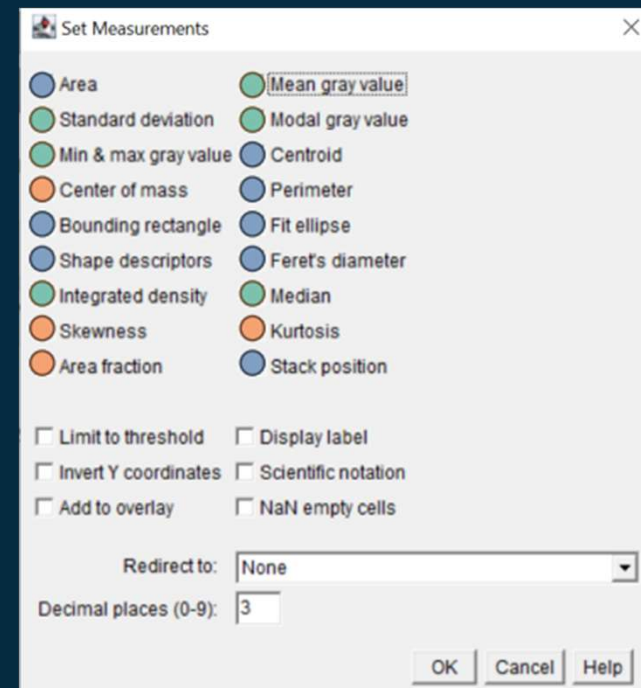


Quantifying Regions (D = 2,3)

Objects represented as an area (2D) or volume (3D)

Analyze > Set Measurements

- Spatial Descriptors
- Intensity Descriptors
- Spatial Distribution of Intensity



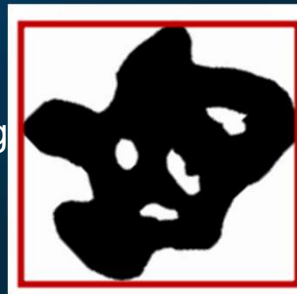
Quantifying Regions (D = 2,3)

Morphology (Shape)

Centroid



Bounding Box



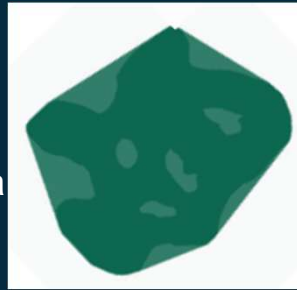
Perimeter



Area



Convex Hull Area



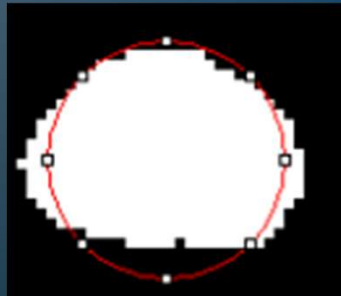
Fit Ellipse
(Major and
Minor axis,
Angle)



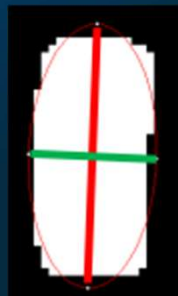
Quantifying Regions (D = 2,3)

Morphology (Shape)

Under “Shape Descriptors” in ImageJ “Set Measurements”



$$Circularity = \frac{4\pi Area}{Perimeter^2}$$



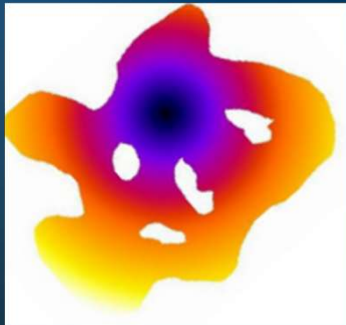
$$Aspect\ Ratio = \frac{Major\ Axis}{Minor\ Axis}$$



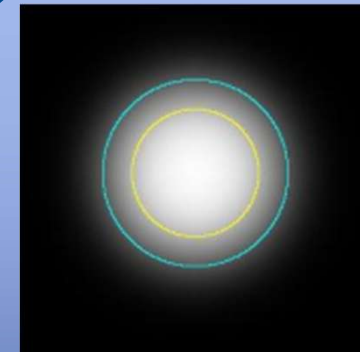
$$Solidity = \frac{Area}{Area\ Convex\ Hull}$$

Quantifying Regions (D = 2,3)

Intensity in object masks



- Min/Max/Median/Mean/Total Intensity values inside the mask
- Center of Mass: Centroid weighted by intensity values



Note: Be careful about segmenting + measuring intensity in the same channel!

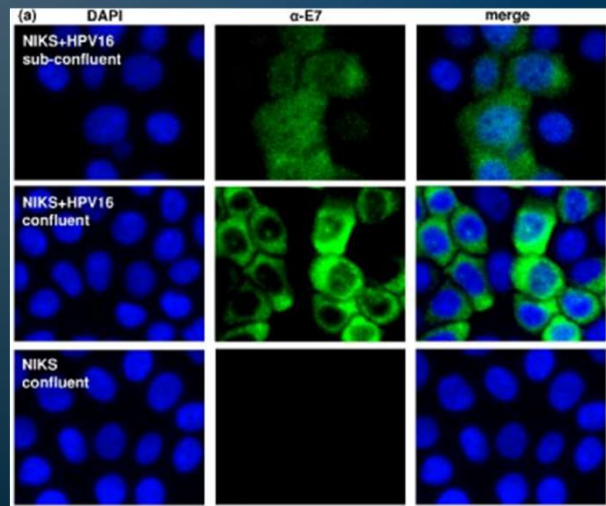
Mean intensity:

- yellow ring = 220
- cyan ring = 175

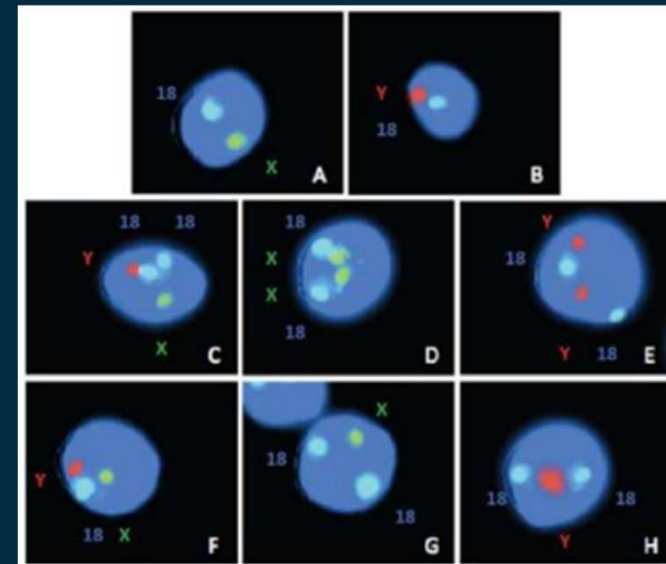
Quantifying Regions (D = 2,3)

Intensity in object masks

- Signal localization (“co-localisation”)
- Object classification



Localisation of Human Papillomavirus 16E7 changes with Cell Confluence

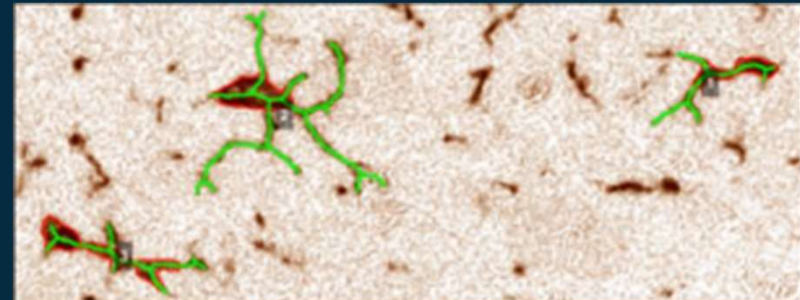


Sample FISH combination: chromosomes X, Y and 18. Healthy spermatozoa (A-B), abnormalities (C-H)

Quantifying Filament and Networks (D = 1)

Objects are represented as curves/lines

- Filaments
 - Lengths
 - Curvature
 - Thickness /with region)
- Networks
 - Branch/junction/end points
 - Loops
 - Tortuosity
 - Fractal Dimensions

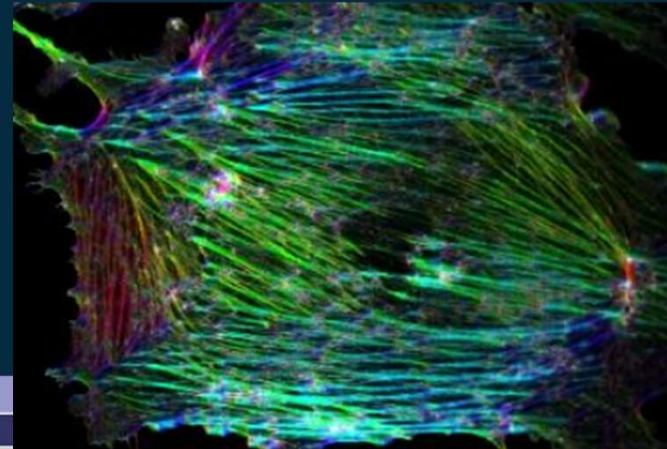


ID	# Branches	# Junctions	# End-points	Average Branch Length
1	7	3	5	31.877
2	19	9	11	35.408
3	11	5	7	23.997

Quantifying Filament and Networks (D = 1)

Considering filaments as a collective
(closer to texture analysis)

- Fiber orientation
- Distribution
- Coherence

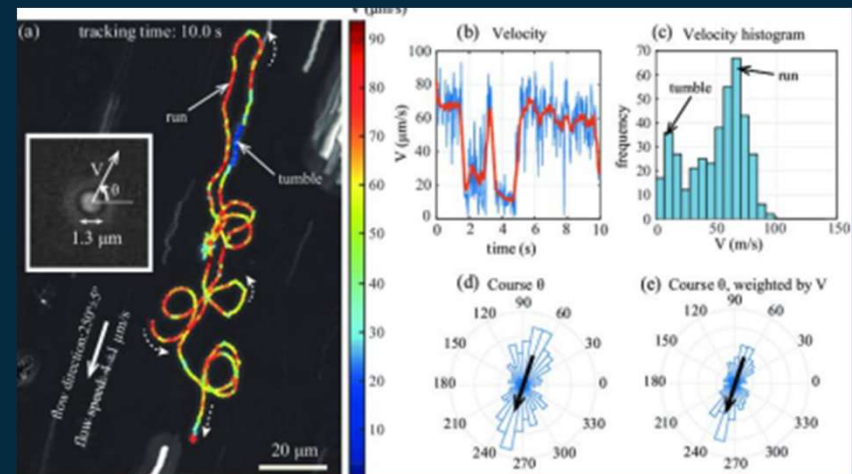


Quantifying Points ($D = 0$)

Objects are represented by their coordinates (position)

- Spatial Coordinates (x, y, z)
 - Density
 - Distance (from other objects)
 - Clustering
- + Temporal Coordinates (t)
 - Tracks
 - Direction
 - Velocity

Can be used in physical modelling

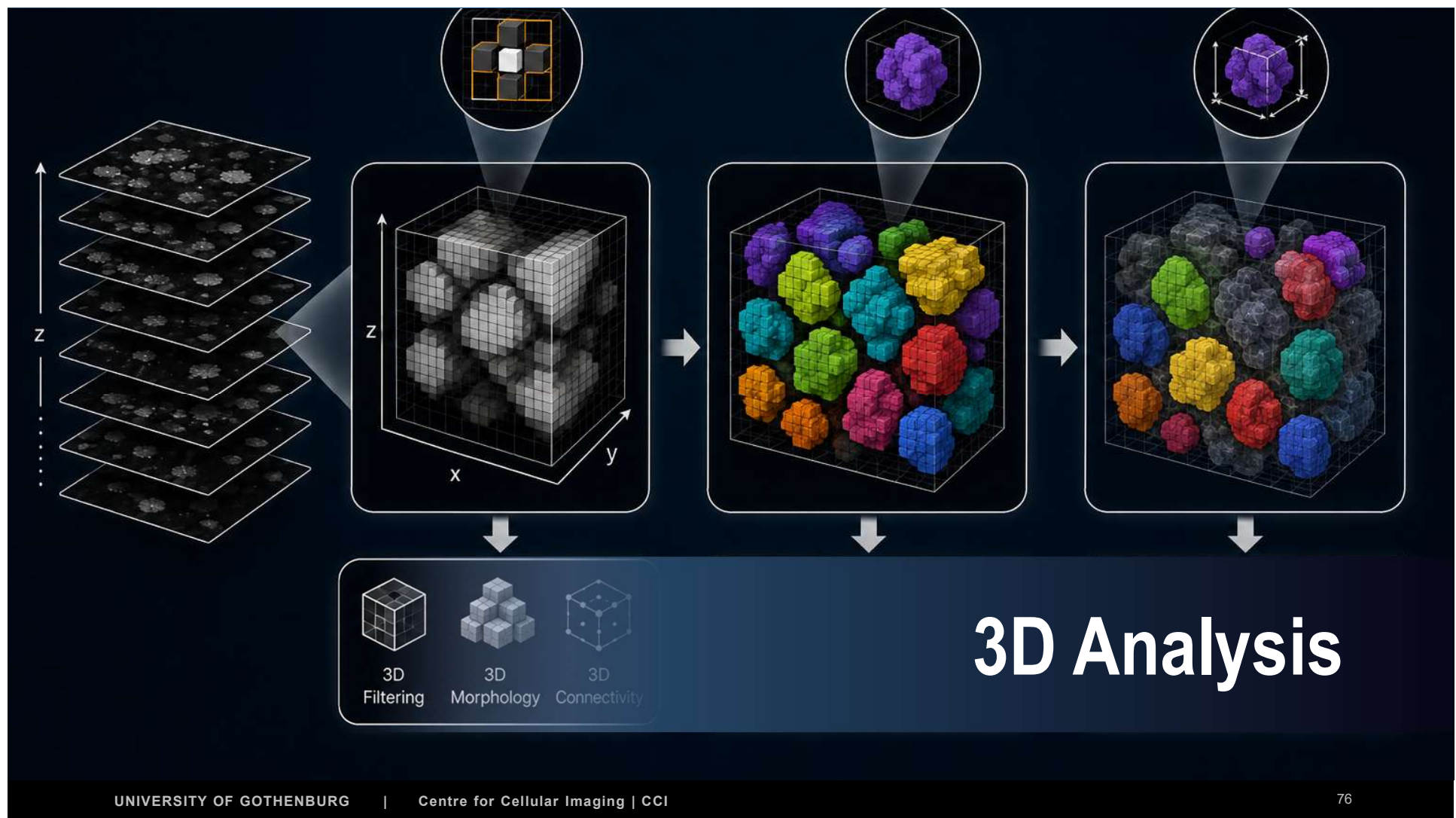


Bacteria track dynamics
Vincent & Kanzo (2017)

Exercise 6: Quantify Objects

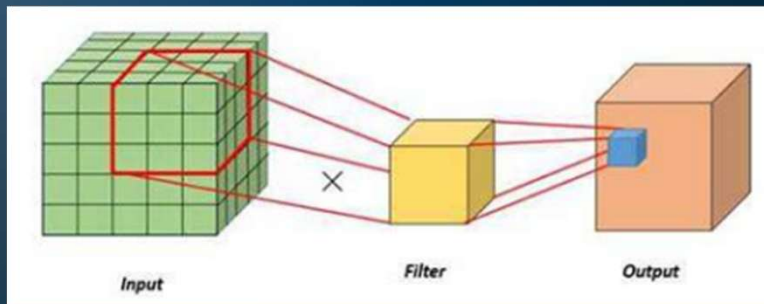
1. Open Image Example 6
2. Globally threshold objects
3. Call **Analyze > Set Measurements**
 1. *Area and Shape Descriptors*
4. Call **Analyze > Analyze Particles**
 1. *Clear results, Display results, and Add to Manager*
5. Inspect values from the column Circ (Circularity) and compare it for the different shapes.
6. Analyze the connected particles again, but this time try to keep only the ellipse like particles by adjusting the circularity filter (in *Analyze Particles*)

Analyze Particles only allows for filtering particles by size and circularity. If you want to use other measurements you will need to write an ImageJ macro to parse the results table and discard invalid objects.

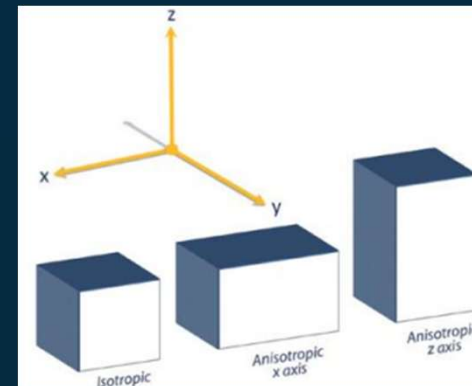


3D Spatial Operations

Filters, morphological operators and quantification can also be performed in 3D on voxels (instead of pixels)



Voxels may not be isotropic (may need to set filters with different sizes in x, y, z)



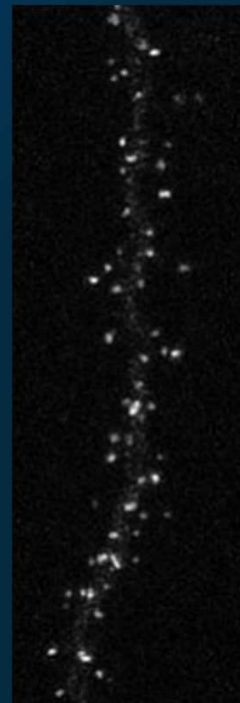
3D Analysis Tools in ImageJ

Some available ImageJ tools

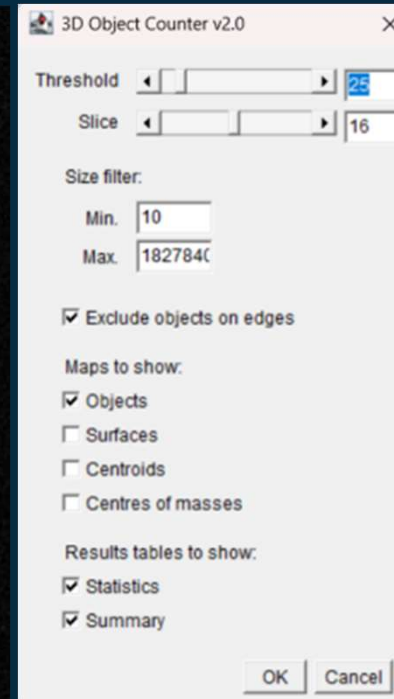
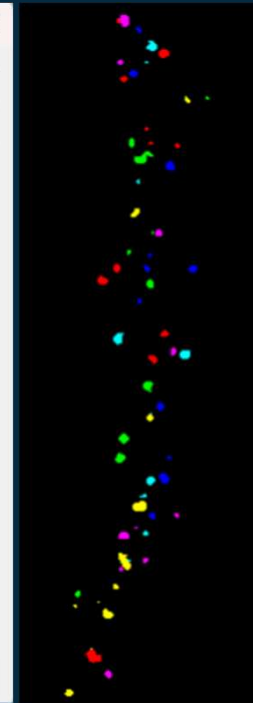
- Analyze > 3D Object Counter
- Plugins > 3D Viewer
- Plugins > BigDataViewer

Useful for image registration
and large datasets

Original Z-Stack

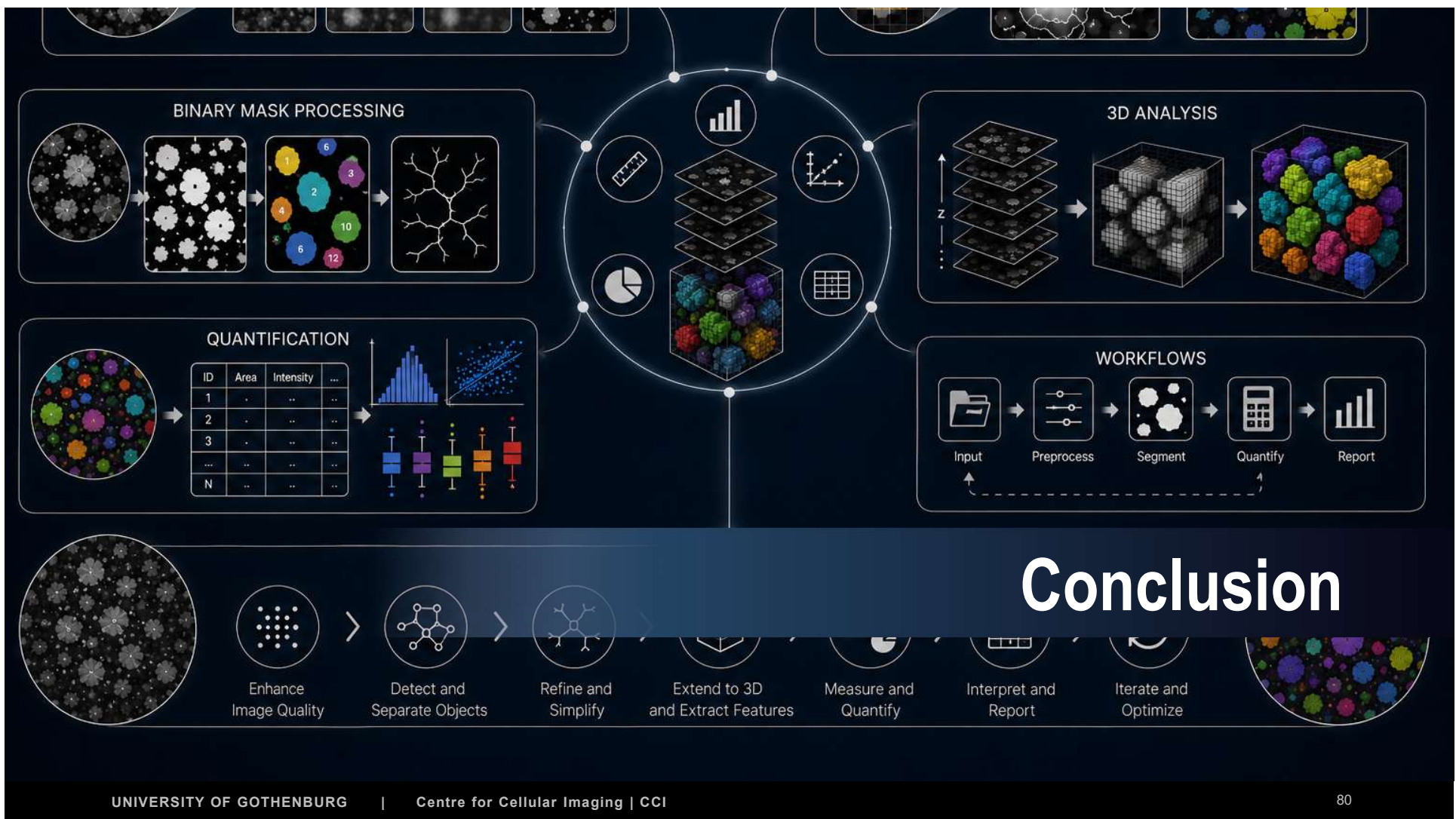


Label Mask

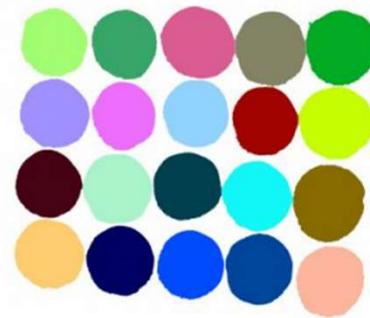
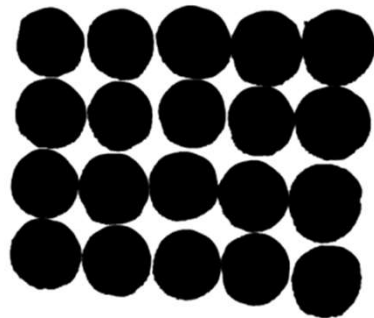


Exercise 7: Quantify Objects

1. Open Image Example 7
2. Call **Process > Filters > Median 3D...** Set all radius values to 2 pixel
3. Threshold the image
 1. *Black background (of binary mask)*
 2. *Stack Histogram*
4. Convert image from binary mask to label mask with **Plugins > MorphoLibJ > Binary Images > Connected Component Labelling**
5. Call **Plugins > MorphoLibJ > Analyze > Analyze Regions 3D**
 1. *Volume, Centroid*



General Image Analysis Workflow



Number of
tomatoes: 20
Average tomato
diameter: 6cm

Image



Segmentation



Objects



Measurements

Image processing



Segmentation



Quantification

Some Key Takeaways

- **Image processing**

- Thinking of digital images as mathematical objects that can be manipulated.
- Spatial context and object models are key to robust image analysis.

- **Image analysis workflows**

- Trade-off between accuracy and generalizability
- Multiple processing steps may be taken to get end results, with specific **methods** (algorithms) and **parameters** used for each steps
 - Importance of documenting your workflow for reproducibility.
 - Coding can help you execute these workflows repeatedly
- When building workflow, consider it in the context of your experimental/research goals

Where to found out More?

ImageJ specific

- [ImageJ User Guide](#) and [ImageJ Wiki](#)

Filters and spatial operations

- [Introduction to Bioimage Analysis – Pete Bankhead](#)
- Examples — [skimage documentation](#)

Other Resources

- Practical Workflow Building Textbook: [Bioimage Data Analysis Workflows – Miura & Sladoj \(Eds.\)](#)
- Image Analysis Forum: [Image.sc Forum](#)

Bonus: BIA-Bubbles Game



Github, [haesleinhuepf/bia-bubbles](https://github.com/haesleinhuepf/bia-bubbles)



UNIVERSITY OF
GOTHENBURG

Thanks for your attention